Navigating the Commons

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Inefficiencies in the commons arise not only from resource use by existing participants but also from their capacity investment and the entry of new firms. This paper develops a model of firm dynamics with common-pool externalities and estimates it using firm-level panel data from the American whaling industry (1804–1909), an unregulated global commons. During the industry's growth, harvesting one more whale imposed an external cost worth about 45 percent of the whale's market price, and two-thirds of that cost came from entry, exit, and investment. Because open access left it unpriced, firms built 3.6 times the first-best vessel capacity, dissipating 24 percent of first-best welfare. Faster technological progress and longer-lasting demand lower welfare under open access, while they raise it under the first best.

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"...whether Leviathan can long endure so wide a chase, and so remorseless a havoc." — Herman Melville, *Moby-Dick*

1. Introduction

This paper investigates how firm entry, exit, and investment shape the tragedy of the commons by analyzing the 19th-century American whaling industry. Commons problems arise when one agent's use of a shared resource reduces others' returns, as in overfishing, deforestation, or groundwater overdraft (Hardin 1968; Ostrom 1990; Stavins 2011; Libecap 2024). While much research focuses on resource use by existing firms, little is known about their investment in production capacity and the entry of new firms. These firm-level dynamics determine how a shared resource is allocated in the long run, and ignoring them may understate the cost of externalities.

Studying firm dynamics in the commons presents three main challenges. First, it requires long panel data that link firms' entry, exit, and investment to resource dynamics, yet such datasets are rarely available in open-access environments. Second, firm behavior and externalities are connected through complex feedback loops. Each firm's actions affect stock depletion and congestion, and these effects in turn shape future decisions, complicating empirical analysis. Third, measuring externalities requires accounting for both contemporaneous and future effects across all affected agents. Doing so is especially demanding as industry structure evolves, continually changing how externalities propagate over time.

To address these challenges, I first construct a firm-level panel of American whaling, tracing its evolution over a century under open access. I then develop a dynamic model in which firms enter, exit, and invest in vessels. Firms are heterogeneous in vessel capacity and productivity, making decisions based on their own state and a few aggregates that summarize externalities. This structure makes the dynamic feedback between firm actions and externalities tractable. Finally, I solve the social planner's problem to recover the shadow value of a whale (and with it the marginal external cost of harvesting) and quantify the cost of open access.

Three main findings emerge from the estimates and counterfactuals. First, the external cost of harvesting a whale was a large fraction of its market price, rising from 20 percent in 1815 to 60 percent in the 1840s. A whale taken today lowers the catch of every voyage that sails later, and how many sail depends on which firms enter, which exit, and how much capacity they build. Those decisions account for two-thirds of the external cost in the early boom, so a valuation that holds the industry fixed understates what a whale is worth to society while the fleet is still growing.

Second, open access overbuilt capacity and dissipated 24% of first-best welfare. At its peak, the industry held 3.6 times the vessel capacity the first best would ever have reached. The extra vessels drew down the whale stock and crowded the hunting grounds, so harvesting efficiency fell by 45% between the 1820s and the 1850s. On the other hand, under the first best, harvesting efficiency would have risen by 31% over the same years. Measured against the first-best allocation, nearly three-quarters of the vessel tonnage the industry built was not worth building.

Third, better technology and longer-lasting demand are good news for any industry, yet under open access both lower welfare. A higher return attracts firm entry and vessel investment whose external cost no one internalizes. Doubling the growth rate of whaling technology would have raised the loss from 24% to 34% of first-best welfare. Petroleum, discovered in 1859, substituted for whale products and reduced the demand for them. Had that demand persisted, the welfare loss would have been 74% of first-best welfare instead of 24%. In both cases, the loss from greater externalities outweighs the gain from higher profitability.

These findings draw on new data, a model of firm dynamics with externalities, structural estimation of its parameters, and counterfactual analyses. I construct a firm-year panel from the expanded American Offshore Whaling Voyages Database (Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association 2024b), supplemented with *Ship Registers* to fill in missing firm information.[1] The sample covers U.S.-registered voyages from 1804 to 1909 and records vessel tonnage, output volumes, and crew size. I link these firm-year observations to annual whale-stock estimates and product prices, yielding a panel that spans the industry's rise and decline with near-complete coverage of U.S. voyages. At its mid-century peak, when externalities were most severe, American fleets dominated global whaling, accounting for roughly 750 of 900 vessels worldwide (Moment 1957). This setting provides a rare opportunity to study a large-scale common-pool resource in the absence of regulation.

The model combines common-pool externalities with competitive industry dynamics. Whaling firms make entry, exit, and investment decisions based on their vessel capacity and productivity (Hopenhayn 1992; Ericson and Pakes 1995).[2] Their actions generate externalities they do not internalize: a stock externality, as overexploitation reduces future resource availability (Gordon 1954), and a congestion externality, as contemporaneous

[1] I use "firms" and "whaling agents" interchangeably. Section 2.3 details the data construction.

[2] Productivity is a key source of heterogeneity in whaling. Firm-level panel data show wide dispersion in output. Some firms harvest only a few whales in a year, while others catch more than a hundred (Table 1). Observable characteristics such as vessel capacity and crew size cannot fully explain this persistent, systematic variation.

activity crowds the hunting grounds (Smith 1969). I model production as a function of firms' own states and two aggregates that summarize the stock and congestion externalities. Firms' actions jointly determine these aggregates, which in turn feed back into individual incentives (Huang and Smith 2014). This structure keeps equilibrium computation tractable while capturing externalities in firms' payoffs.

I estimate the model in two steps. First, I identify the production function and the demand curve. I estimate the production function using firm-level panel data on physical inputs (vessel capacity and crew size) and output (whales harvested). To address endogeneity between productivity and input choices, I employ a standard dynamic panel approach that models productivity as an AR(1) process and uses lagged inputs as instruments. I estimate the demand curve using supply-side cost shifters as instruments to recover the price elasticity consistently. Instruments for price include whale stocks five years earlier and long-term bond yields in New England, both of which affect whaling costs but not demand directly.

In the second step, I estimate dynamic costs for entry, exit, and capacity adjustment using a full-solution approach. Estimating these costs requires computing firms' optimal decision rules for both discrete (entry and exit) and continuous (capacity adjustment) decisions, which can be sensitive to the discretization of the continuous state and action spaces. I therefore adopt the recent methodology of Gowrisankaran and Schmidt-Dengler (2025). This approach introduces random shocks to marginal adjustment costs, making the implied choice probabilities robust to the discretization. I then use the implied equilibrium choice probabilities to estimate the dynamic cost parameters that best rationalize firms' observed decisions in the data.

For the counterfactuals, I solve for the allocation a social planner would choose. The planner decides how many firms enter, how many exit, and how much vessel capacity they hold, accounting for the social value of the whales they harvest. A whale left in the ocean reproduces, and it raises what every later voyage harvests, because a larger whale stock makes every firm more productive. That value is the shadow value of a whale, and it is the cost a firm imposes on everyone else when it takes one. The shadow value therefore depends on entry, exit, and investment, which determine how many firms and vessels will be hunting in later years. The difference between the market outcome and the planner's allocation is what open access cost.

Relationship to the literature. The main contribution of this paper is to show how firm-level dynamic decisions shape the long-run allocation of common-pool resources.

My model builds on recent advances in the study of common-pool resources and industry dynamics. A seminal contribution is Huang and Smith (2014), who examine dynamic strategic interactions in shrimp fisheries. The central challenge in their setting is that agents' payoffs depend on others' actions through dynamic externalities, complicating empirical analysis. To address this issue, they assume that fishers affect others only through the aggregate number of vessels, which in turn governs shrimp stocks. Their model features a fixed set of players, which is appropriate over relatively short horizons. Over longer horizons, especially under open access, firms enter, exit, and invest. A large literature on industry dynamics shows that these firm-level margins shape industry structure and resource allocation (see Aguirregabiria, Collard-Wexler, and Ryan 2021 for a review). I connect these strands by embedding entry, exit, and capacity adjustment into a common-pool industry equilibrium. This perspective highlights how entry and investment can outpace exit and divestment, generating persistent inefficiency through overcapacity.

This paper also relates to a growing literature on management institutions in fisheries. Costello, Gaines, and Lynham (2008) show that catch shares sustain global fish stocks, and Costello et al. (2016) compare overfishing outcomes across management regimes. Researchers have also examined policy effects beyond stock levels. Ho (2023) studies productivity gains from individual transferable quotas. Englander (2023) identifies temporal and spatial spillovers from temporary closures. Aspelund (2025) analyzes redistribution and efficiency effects of trade restrictions in fishing-permit markets. Segerson and Squires (1993) highlight capacity choices, showing that output quotas can induce divestment, and a theoretical literature identifies capital as the important margin, because vessels are durable and their capacity stays in the fishery once built (Clark, Clarke, and Munro 1979; Squires et al. 2010). However, direct evidence on the welfare effects of firm dynamics in the commons remains scarce. I directly model firms' dynamic decisions as determinants of long-run efficiency in commons regulation, and I quantify the dynamic costs of open access over a full industry life cycle.

A related literature shows how economic environments shape the history of shared resources. Brander and Taylor (1998) model Easter Island's collapse as a renewable-resource feedback, and Taylor (2011) attributes the near-extermination of the American bison to international trade. On the theory side, Copeland and Taylor (2009) show how technology, institutions, and prices shape commons outcomes, and Noack and Costello (2024) trace overexploitation to credit expansion without secure property rights. I find that the substitution of petroleum for whale products reduced the cost of open access, while advances in whaling technology raised it. What looked disastrous for the industry helped to alleviate the commons problem, whereas what drove the industry's prosperity exacerbated the tragedy.

Beyond the commons, a growing literature studies regulation design in environmental and energy markets with industry dynamics. Ryan (2012) and Fowlie, Reguant, and Ryan (2016) examine market-based environmental regulation under market power and emissions leakage. Elliott (2024) investigates carbon taxes and capacity subsidies when emissions trade off against intermittency from wind and solar. Aronoff and Rafey (2023) study welfare gains and externalities from environmental offsets. Butters, Dorsey, and Gowrisankaran (2025) analyze the equilibrium effects of large-scale battery storage. Gowrisankaran, Langer, and Zhang (2025) and Chen (2024) show that policy uncertainty shapes dynamic market efficiency. Relative to this literature, I develop an empirical framework for firm dynamics in the commons, in an environment where the externalities complicate the dynamic problem. The framework of this paper applies to many industries whose central distortion comes from a shared resource.

Finally, this paper speaks to the literature on long-run productivity in economic history, which has traced productivity growth to management practices (Braguinsky et al. 2015; Giorcelli 2019; Bianchi and Giorcelli 2022; Rubens 2023), to technology adoption and innovation (Hanlon 2015; Braguinsky et al. 2021; Juhász, Squicciarini, and Voigtländer 2024; Hornbeck et al. 2025), and to industrial policy (Juhász 2018; Lane 2025). In American whaling, harvesting efficiency falls through the industry's growth even as the whaling technology improves, because the resource (whales) every firm shares is being drawn down. A decades-long productivity decline in a shared-resource industry is a unique property of the unmanaged institution, and the counterfactuals show how much of that decline management would have restored.

Outline. The rest of the paper is organized as follows. Section 2 provides a historical overview and descriptive analyses of the American whaling industry. Section 3 develops the model of common-pool industry dynamics. Section 4 describes the estimation strategies and presents the results. Section 5 implements counterfactual exercises. Section 6 concludes.

2. The American whaling industry

This section provides a historical overview of the US whaling industry and describes the data. I document the industry's major products and the shifts in market size (Section 2.1), the primary decision-makers (Section 2.2), data sources (Section 2.3), and descriptive statistics (Section 2.4).

2.1. Products and markets

The American whaling industry primarily produced three commodities: sperm oil, whale oil, and whalebone (baleen).[3] Sperm oil, rendered by boiling the blubber of sperm whales, was prized for its clean, bright flame and was used for illumination and as a lubricant for fast-moving machinery (e.g., spindles). Whale oil, extracted from baleen whales, was used as a lubricant for heavy machinery and as an illuminant. Whalebone, taken from the mouths of baleen whales, was used in products such as corsets. Sperm whales were hunted on warm-water grounds across all oceans, while the baleen species, chiefly right and bowhead whales, concentrated in higher latitudes (Tillman and Donovan 1983). Processing took place on board. Blubber was rendered in tryworks, brick furnaces built into the deck, which let vessels stow oil and remain at sea for years. The product market consisted of homogeneous goods and was competitive. Individual whaling firms had little ability to affect prices or differentiate their outputs from those of other firms (Davis, Gallman, and Gleiter 2007).

During the first half of the nineteenth century, industrialization in the United States substantially increased demand for oil, expanding markets for whaling. During this period, the United States overtook Britain as the leading whaling nation, supported by strong domestic demand growth, technological and institutional advantages, and higher-skilled crews (Davis, Gallman, and Hutchins 1987). The industry entered its "Golden Age" during the 1830s–50s, when American whalers produced over 90% of the world's sperm oil (Tillman and Donovan 1983).[4] The mid-1840s brought a series of shocks. Coal oil emerged as a credible substitute, the California gold rush drew away crews, and voyages shifted increasingly toward Pacific grounds. Thereafter, the discovery of petroleum in 1859 and the disruption of the Civil War (1861–1865) prompted the decline of the industry.[5] By the late nineteenth century, output and value had fallen back to levels comparable to those of the early 1800s.[6]

[3] Online Appendix Figure F1 shows trends in output quantities and prices.

[4] I treat non-American whaling operations as exogenous given their relatively small scale. If they responded endogenously to counterfactual regulations on American whaling but were not themselves regulated, the effects of the planner's management in Section 5 would be smaller.

[5] Whalers understood the threat quickly. The *Whalemen's Shipping List and Merchants' Transcript* (WSL, July 2, 1861) reported:

> At New Bedford the business has declined about one-third during the past three years, and it is believed will decline fully a third more within the present year [1861] ... Oil, which costs 60 cents to produce, will now bring but 40 cents.

Likewise, Tower (1907) dates the fate of the whale fishery to the opening of the first Pennsylvania oil well.

[6] International regulation of whaling began only with the 1931 Geneva Convention, well after the American

2.2. Whaling agents

Whaling agents, interchangeably referred to as "firms" in this paper, were the primary decision makers (see Chapter 10 of Davis, Gallman, and Gleiter 2007 and Chapter 1 of Nicholas 2019 for a more in-depth discussion of whaling agents). The median firm ran two vessels in a given year, and roughly half of firm-year observations involve a single vessel (Table 1). As American whaling voyages spanned multiple seasons (typically two to three years), success depended on effective planning and management by agents. They were responsible for acquiring vessels and hiring a captain, as well as determining the necessary equipment and crew. In the process of decision making, agents utilized logbooks from prior voyages. These contained daily notes made by captains and first mates, providing detailed information on whale sightings, weather, locations, and crew morale. At the end of a venture, the logbook became the property of the agent, serving as a repository of accumulated knowledge (Nicholas 2019).

Plans by agents included voyage duration, hunting grounds, locations and dates for resupplying, and shipping oil or bone back home (see Online Appendix B.1 for an example). The plan was made before the vessel sailed but was frequently subject to change. Unforeseen success or challenges could necessitate altering the original schedule. For instance, exceptional success might prompt an early stop at a transshipment point; setbacks could extend the time at sea beyond the initial plan. While the captain made day-to-day sailing and hunting decisions, the agents held general authority, ensuring consistent communication between them as best they could. Before the voyage commenced, the captain and agent agreed on dates and stations for exchanging letters. These exchanges occurred at predetermined rendezvous points, either between whaling vessels or between supply ships and whaling vessels (see Online Appendix B.2 for an example).[7]

In many cases, whaling agents owned vessels as principal owners, flying a "house flag" that signaled ownership. But this did not mean the agent alone financed the voyage. Multiple investors typically contributed, with arrangements varying from one voyage to another (see Online Appendix Figure C2 for an example). The reliance on multiple investors reflected the substantial capital required to outfit a whaling expedition. A typical New Bedford voyage in the 1850s cost between \$20,000 and \$30,000 (equivalent to roughly \$831,000–\$1,200,000 today). The cost was considerable, especially when compared to other sectors. The value of the average manufacturing firm's capital stock was about \$4,335 in

industry had disappeared (Tønnessen and Johnsen 1982).

[7] For instance, Santa Maria, a small island in the Galapagos archipelago, became a mail hub because it was frequented by vessels hunting whales in the Pacific (Nicholas 2019).

1850 (≈\$180,000 today) and \$7,191 in 1860 (≈\$285,000 today) (Davis, Gallman, and Gleiter 2007).

2.3. Data

I construct several datasets for examining the whaling market. This subsection describes the new aspects of the data, emphasizing how they reveal (i) firm-level behaviors; (ii) price per whale; (iii) common-pool features; and (iv) demand shifters. See Online Appendix C for a detailed description.

The primary source is the American Offshore Whaling Voyages Database (*Voyage Database*), which covers nearly all U.S. whaling voyages undertaken in the 19th century (Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association 2024b).[8] The database records vessel characteristics (name, unique ID, tonnage, rigging), the managing owner/agent, captains, outbound and inbound years, outputs (sperm oil, whale oil, whalebone), ports of departure and return, and other related information (see Online Appendix Figure C1 for an example).

The *Voyage Database* does not include crew information, so I supplement it with the American Offshore Whaling Crew Lists Database (*Crew Lists Database*) to obtain crew size for each voyage (Mystic Seaport Museum, New Bedford Whaling Museum, and Nantucket Historical Association 2024a). The *Crew Lists Database* covers roughly half of the voyages in the *Voyage Database*.

To study firms' production, entry, exit, and capital accumulation, I build a firm-year panel from voyage records. This construction becomes possible because the *Voyage Database* recently expanded whaling agent coverage from about one-fifth to nearly two-thirds of the fifteen thousand voyages.[9] Still, the *Voyage Database* lists no whaling agent for 5,182 of the 15,707 voyages, so I further supplement agent information by manually checking the data and reading port *Ship Registers* (see Online Appendix C.2). Together these raise agent coverage to 12,612, or about 80 percent of all voyages.

Based on this expanded dataset, I divide each voyage's output by its duration (measured from departure to return) and aggregate to the firm-year level by whaling agent. I convert the three outputs (sperm oil, whale oil, and whalebone) into the *number of whales harvested* using historical benchmarks (see Online Appendix C.3). I also aggregate voyage's inputs

[8] Prior studies in economics use related data sets to study risk sharing in financial markets, firm formation and corporate organization, productivity, and labor diversity (Hilt 2006, 2007, 2008; Davis, Gallman, and Gleiter 2007; Baggio and Cosgel 2024).

[9] Earlier studies relied on substantially more limited coverage (e.g., Hilt 2006; Hilt 2007; Hilt 2008; Davis, Gallman, and Gleiter 2007)

(vessel tonnage and crew size) to the firm–year level, assuming they remain constant throughout each voyage.[10]

The prices of whaling products come from Tower (1907) and Davis, Gallman, and Gleiter (2007). I calculate the price *per whale* as the aggregate value of outputs divided by the total number of whales harvested each year. The aggregate output value equals sperm oil quantity × sperm oil price + whale oil quantity × whale oil price + whalebone quantity × whalebone price. I compute the total number of whales harvested by summing catch records across all voyages. Online Appendix Figure F2 reports the annual whale price.

Two aggregate series complete the panel. Total vessel tonnage, summed over all whaling vessels in each year from the *Voyage Database*, measures fleet-wide capacity. The whale stock series is projected from whale population estimates (Davis, Gallman, and Hutchins 1988; Whitehead 2002) and catch data using a discrete-time generalized logistic model, described in Section 3.1.

Finally, I use two exogenous variables to capture demand shifters: U.S. GDP per capita and petroleum prices. These data come from the Historical Statistics of the United States (Carter et al. 2006).[11]

2.4. Descriptive statistics

Table 1 reports firm-year statistics that highlight substantial variation across observations. Panel A shows that some firms harvested only a few whales, while others harvested more than one hundred. Panel B shows wide dispersion in operational scale. Annual operations ranged from a single vessel to more than seven, with corresponding variation in vessel tonnage and crew size. Firm lifetimes also varied substantially. Some firms operated only briefly, while others remained in the industry for more than 30 years. The median firm spent six years in the industry, two to three voyages at typical voyage lengths.[12] Panel C summarizes dynamic decisions. Entry is defined as the first appearance as a whaling agent in the dataset, and exit means leaving whaling altogether. Capacity adjustment occurred at the vessel margin: firms invested by acquiring additional or larger vessels and divested by selling or retiring them. Investment and divestment are measured as changes in vessel tonnage from year t to year $t+1$, excluding changes due to wreckage.[13] On average, relative

[10] Although crew desertion and recruitment occurred, vessel tonnage largely constrained crew size.

[11] The online version is available at https://hsus.cambridge.org/. Annual U.S. GDP per capita is from Series Ca9, and petroleum prices are from Series Db56. Online Appendix Table F1 reports summary statistics.

[12] The firm-age row of Table 1 reports age at each firm-year observation, whose median (7 years) exceeds the median firm's six-year span because long-lived firms contribute more observations.

[13] Shipwrecks were a constant hazard. Roughly nine vessels per year were destroyed over the sample period, with spikes during the Civil War and the Arctic disaster of 1871; about 6% of firm-year observations

TABLE 1. Firm-year level statistics for the American whaling industry

Variable	Unit	Mean	SD	P5	Median	P95
Panel A. Outputs						
Whales harvested	Number	39.896	48.480	1.91	22.56	139.61
Sperm oil	Thousand barrels	0.502	0.694	0.00	0.28	1.82
Whale oil	Thousand barrels	0.766	1.286	0.00	0.22	3.48
Whale bone	Thousand pounds	6.596	13.697	0.00	0.13	32.55
Panel B. Inputs						
Vessel capacity	Tonnage	737.128	792.062	106	383	2,452
	Number	2.559	2.376	1	2	7
Crew size	Number	82.760	84.098	16	50	262
Firm age	Year	11.250	10.904	1	7	33
Panel C. Dynamic decisions						
Entry	Rate	0.110	0.098	0.000	0.098	0.311
Exit	Rate	0.108	0.053	0.020	0.105	0.217
Investment	Rate	0.144	0.107	0.035	0.132	0.245
	Tonnage	326.306	212.755	72.4	311.5	747.0
Divestment	Rate	0.138	0.081	0.000	0.132	0.281
	Tonnage	-343.067	239.686	-819.1	-313.0	-75.0

Note: The dataset includes 11,176 firm-year observations and 1,126 firms, covering the years 1804 to 1909. Reported quantities of sperm oil, whale oil, and whalebone determine the number of harvested whales (see Online Appendix C.3 for details). Crew size is observed for 6,103 firm-year observations because the *Crew Lists Database* is a subsample of the *Voyage Database*. Entry and exit rates are defined as the fraction of firms that enter or exit in a given year; investment and divestment rates are defined analogously. Changes in vessel tonnage between the current year and the following year capture investment and divestment, excluding changes due to wreckage.

to the number of firms active in each year, 11% entered, 10.8% exited, 14.4% invested, and 13.8% divested. Median investment was about 311 tons and median divestment about 313 tons,[14] with substantial dispersion in both, reflecting the lumpy and uncertain nature of capacity adjustment.

Figure 1 presents the evolution of total whaling vessel capacity and whale stock. Panel A shows the rise of American whaling to global dominance during the industry's Golden Age, followed by its collapse within a single century. In terms of capital stock, the industry employed an annual average of only eighteen thousand vessel-tons in 1816–1819. Over the subsequent three decades, total tonnage increased more than thirteen-fold, peaking at 246,000 tons in 1845. Panel B displays the projected whale population, the sum of the

record a wreck. In the model, I treat wrecks as an exogenous destruction hazard, separate from firms' voluntary exit decisions.

[14]A typical tonnage of a vessel was about 250 to 300.

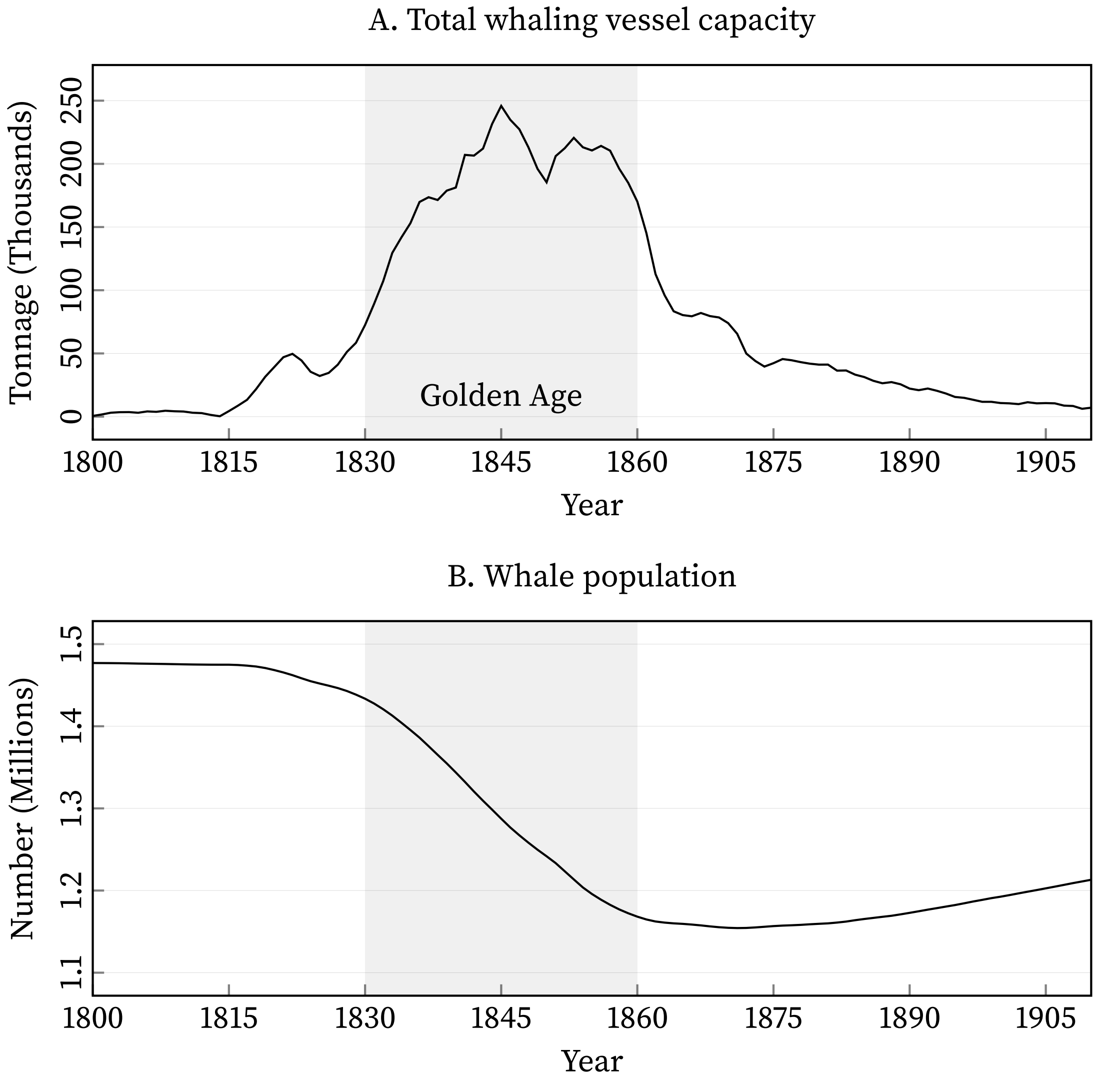

FIGURE 1. Aggregate whaling vessel capacity and whale stock

Note: The term "Golden Age" refers to the period when the American whaling industry was at its peak, from the 1830s to the 1850s. Total whaling vessel capacity is calculated by summing the tonnage of all whaling vessels for each year from the *Voyage Database*. Whale population is projected using a discrete time, generalized logistic model of whale population dynamics (see Eq. 1 in Section 3.1).

baleen and sperm whale stocks (see Section 3.1).[15] Whale stocks experienced sharp declines during the Golden Age, coinciding with the rapid expansion of vessel capacity and hunting pressure.

Figure 2 provides suggestive evidence of negative externalities. Panel A displays the number of whales harvested per thousand tons of vessel capacity. Following a marked

[15] I treat the stock as a single global pool. Whales migrate across ocean basins, the fleet hunted the major grounds worldwide, and no ground-level stock series exists for the period.

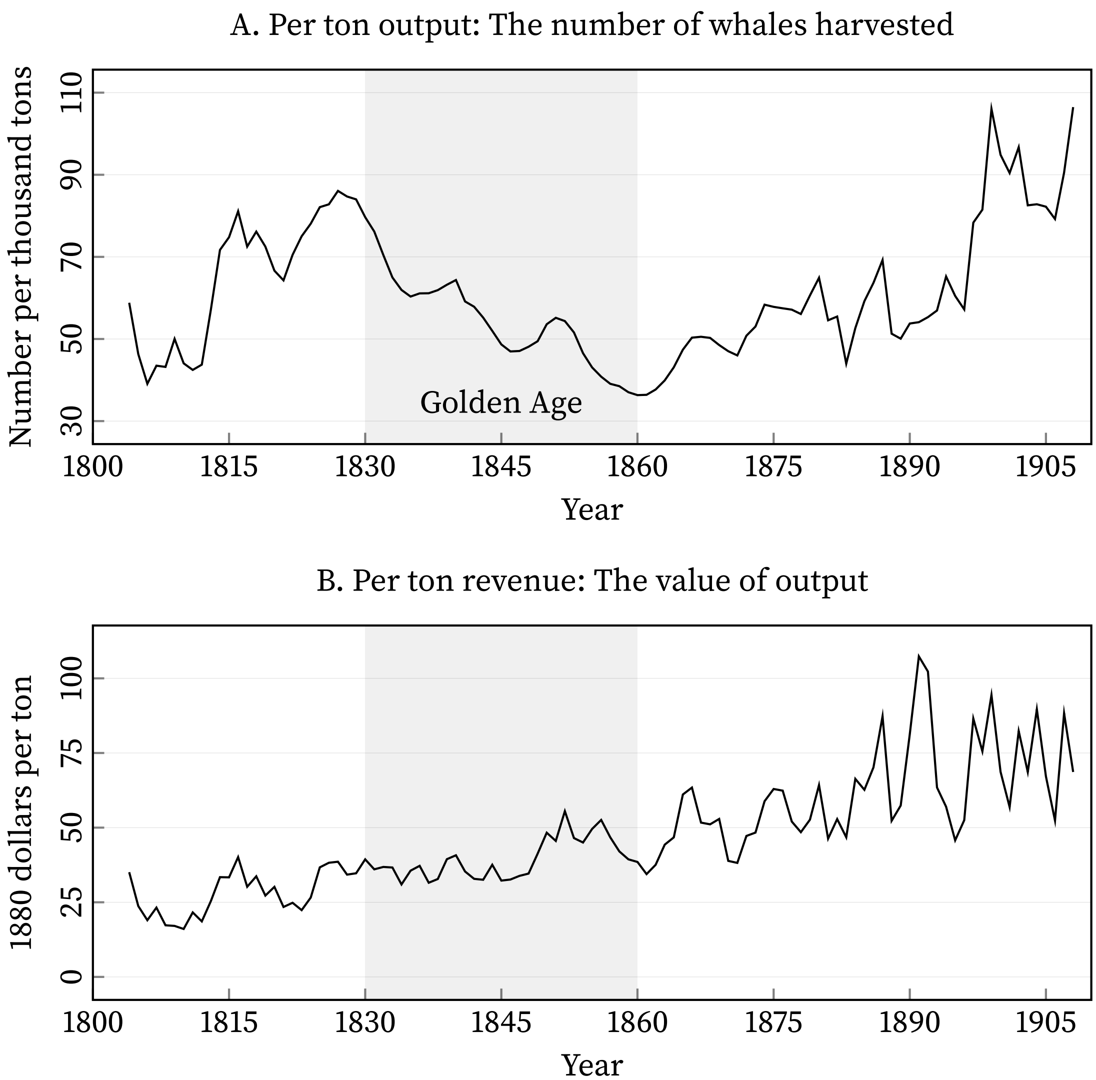

FIGURE 2. Suggestive evidence of negative externalities

Note: In panel A, output is measured by the number of whales harvested per thousand tons of aggregate vessel capacity. In panel B, the value of output per ton of aggregate vessel capacity is calculated as the sum of the product values: sperm oil quantity × sperm oil price + whale oil quantity × whale oil price + whalebone quantity × whalebone price. The term "Golden Age" refers to the period when the American whaling industry was at its peak, from the 1830s to the 1850s.

increase in the 1800s–1820s, it peaked at 86 in 1827 and fell to 43 by the mid-1850s. This decline is striking in light of technological growth over the period, about 1.3% per year from the production function estimates (see Section 4.1). On the other hand, Panel B shows non-decreasing revenue per ton of vessel capacity. It averaged $31 per ton (in 1880 dollars) in the 1820s and rose to $47 in the 1850s. A significant rise in prices resulted in this increase. For instance, the price of sperm oil averaged $0.53 per gallon (in 1880 dollars) in the 1820s and nearly tripled by the 1850s (see Online Appendix Figure F1). These price increases sustained firms' incentives to continue operating despite rising costs, reinforcing the equilibrium effect of continued entry and investment amid worsening common-pool conditions.

3. A model of common-pool industry dynamics

This section develops a model of competitive industry dynamics for a common-pool resource. The model places a large number of price-taking firms (Hopenhayn 1992) in an environment with an endogenous, evolving production externality. It draws on tractable solution methods for industry dynamics with many firms (Weintraub, Benkard, and Van Roy 2008; Benkard, Jeziorski, and Weintraub 2024). The subsections present the model setup (Section 3.1), industry states, beliefs, and decision rules (Section 3.2), dynamic optimization (Section 3.3), and the equilibrium (Section 3.4).

3.1. Setup

Time. Each period corresponds to one year and is indexed by $t \in \{1, 2, \ldots, T\} \subset \mathbb{N}$, where T denotes the terminal year. A finite horizon is crucial for the empirical analysis because it allows the model to capture the industry's rise and fall in a nonstationary environment. By contrast, the infinite-horizon stationary framework does not naturally accommodate this historical pattern.

Firms. Firms are indexed by i. In year t, the set of incumbents is $\mathcal{I}_t$, with $|\mathcal{I}_t| = N_t$, and the set of potential entrants is $\mathcal{I}_t^{\text{pe}}$, with $|\mathcal{I}_t^{\text{pe}}| = N_t^{\text{pe}}$.[16]

Resource. The whale stock in year t is denoted by $W_t \in [0, W_1]$, where W_1 is the natural stock level before the large-scale harvesting by American whalers.[17] The stock evolves according to a discrete-time generalized logistic model of population dynamics (Whitehead

[16]The pool size of potential entrants N_t^{pe} is calibrated to the industry's history: 20 potential entrants per year through 1815, growth of 5% per year during the postwar expansion (reaching about 65 by 1840), a 3% annual decline thereafter, and a 5% annual decline after the discovery of petroleum. The pool always exceeds the number of entrants observed in the data. Though alternative specifications of the potential-entrant pool affect dynamic cost estimates, the findings from counterfactuals are robust.

[17]$W_1 \approx 1.48$ million whales, comprising about 1.11 million sperm whales (Whitehead 2002) and about 367,000 baleen whales (Davis, Gallman, and Hutchins 1988). I project the two populations forward from these pre-exploitation levels under the generalized-logistic law (1), net of recorded catch, to obtain the stock series in Figure 1.

2002; Baker and Clapham 2004; Breiwick and York 2009), net of the year's harvest:

$$W_{t+1} = \left(W_t + rW_t \left[1 - \left(\frac{W_t}{W_1} \right)^z \right] - Q_t \right) \phi^W_{t+1}, , \tag{1}$$

where $\log \phi^W_{t+1} \sim \mathcal{N}(\mu_W, \sigma^2_W)$ is i.i.d. across years, r is the whale-stock regeneration rate, z is the exponent setting the maximum sustainable yield level (MSYL),[18] and Q_t is the total whale harvest in year t. Firms decide before ϕ^W_{t+1} realizes, and they lack a contemporaneous assessment of whale abundance, so each firm acts on the distribution of next year's stock rather than on its level. I write $\widetilde{W}_{t+1}$ for the stock under that distribution, whose dispersion is governed by σ_W.[19]

Firm state. Active whaling firms are heterogeneous along two dimensions. First, capacity (K_{it}) captures the tonnage of operating vessels. Second, productivity (Ω_{it}) reflects managerial efficiency, modeled as Hicks-neutral technology.[20] The *individual* state of firm i in year t is denoted by $x_{it} \equiv (K_{it}, \Omega_{it}) \in \mathcal{X}$. Vessel capacity changes through costly adjustment (investment or divestment) I_{it}:

$$K_{it+1} = (1 - \delta)K_{it} + I_{it}, \tag{2}$$

where δ is the vessel depreciation rate. The index of (log) productivity, $\omega_{it} = \log \Omega_{it}$, follows a first-order Markov process:

$$\omega_{it+1} = \mathbb{E}\left[\omega_{it+1} | \omega_{it}\right] + \nu_{it+1}, \tag{3}$$

where ν_{it+1} is the unexpected productivity innovation, with $\mathbb{E}\left[\nu_{it+1} | \omega_{it}\right] = 0$. Potential entrants are identical prior to entry, sharing a common entry state x^{e}.[21]

[18] The MSYL is the stock level at which the maximum number of whales can be taken without changing the stock level.

[19] I set the location μ_W so that the shock leaves expected harvest unchanged.

[20] Incorporating firm-level, time-varying productivity into the state space addresses the limitation of relying solely on i.i.d. random shocks as the source of unobserved variation over time (Bajari, Benkard, and Levin 2007). It is also crucial for capturing dispersion in output even after accounting for observed firm characteristics such as capacity and crew size (see Table 1). The whale stock enters the harvesting function directly, so Ω_{it} measures a firm's efficiency at a given stock level; it does not reflect the abundance of whales.

[21] I use the entrants' capacity and productivity as those of the median entrant observed in the data.

Production. The whale-harvesting function $\mathcal{H}$ gives the number of whales harvested by firm i in year t:

$$Q_{it} = \mathcal{H}_t(x_{it}, W_t, K_t; \beta). \tag{4}$$

The harvesting function incorporates the stock externality through W_t and the congestion externality through aggregate whaling vessel capacity K_t (the more vessels, the more congested the hunting grounds). Thus, firm i's harvest depends not only on its own state (x_{it}) but also on these external factors. The time subscript in $\mathcal{H}_t(\cdot)$ reflects exogenous technological growth in the industry. The vector β contains the parameters governing the harvesting function.

Demand. The inverse whale-demand function $\mathcal{P}$ describes the relationship between price, aggregate harvest, and demand conditions:

$$P_t = \mathcal{P}(Q_t; Z_t^d, \alpha) = \mathcal{P}_t(Q_t; \alpha). \tag{5}$$

Here, P_t denotes the market price in year t, common to all firms because the American whaling industry was competitive (Davis, Gallman, and Gleiter 2007). The vector Z_t^d consists of exogenous demand shifters, namely U.S. GDP per capita, a time trend, and petroleum prices (since 1859). For notational simplicity, these shifters are absorbed into the time subscript of $\mathcal{P}_t(\cdot)$. The vector α contains the parameters of the demand function.

Period payoff. Firm i's profit Π in year t is revenue minus labor costs:

$$\Pi_t(x_{it}, W_t, K_t, Q_t; \alpha, \beta) = \mathcal{P}_t(Q_t; \alpha)\mathcal{H}_t(x_{it}, W_t, K_t; \beta) - w_t^L L_{it}, \tag{6}$$

where L_{it} denotes crew size and w_t^L is the common wage index for crew. The payoff depends on the decisions of all firms, as reflected in W_t, K_t, and Q_t.

Dynamic actions. In each year t, incumbent firm i chooses whether to exit or remain active. If it stays, the firm may adjust its vessel capacity for the next year. Incumbent actions are denoted $a_{it} \in \mathcal{A} = \{0\} \cup \mathcal{K}$, where 0 represents exit and $\mathcal{K}$ is the set of feasible capacity levels. Potential entrants decide whether to enter the market, with actions denoted $a_{it}^{\text{pe}} \in \mathcal{A}^{\text{pe}} = \{0, 1\}$, where 0 indicates no entry and 1 indicates entry.

Timing. At the start of year t, incumbent firms hunt whales and earn period payoffs as defined in equation (6). After receiving payoffs, each incumbent draws an exit value (ζ_{it}), and each potential entrant draws an entry cost (κ_{it}). An incumbent that stays then draws a marginal capacity adjustment cost (ϵ_{it}) before choosing next year's capacity. Firms observe these draws before deciding, so each cost and value shock enters the payoff of the action it prices.[22] At the end of the year, incumbent firms that chose to stay face the wreck hazard. Surviving firms' decisions are implemented, entrants enter, and states transition accordingly.

3.2. Industry states, beliefs, and decision rules

There are many firms, and each is small relative to the industry size. Firms take the market price and the evolution of the industry aggregates as given. They perceive that these aggregates evolve deterministically as idiosyncratic shocks average out across a large number of firms (Weintraub, Benkard, and Van Roy 2008, 2010).[23] Each firm therefore conditions its decisions on its own state, the whale stock, and the industry aggregates.

As shown in equation (6), the industry aggregates that affect firms' payoffs are total vessel capacity and total whale harvest. Total vessel capacity equals the sum of all firms' capacities:

$$K_t = \sum_{i \in \mathcal{I}_t} K_{it}. \tag{7}$$

Total whale harvest equals the sum of all firms' harvests:

$$Q_t = \mathcal{Q}_t(W_t, K_t; \beta) = \sum_{i \in \mathcal{I}_t} \mathcal{H}_t(x_{it}, W_t, K_t; \beta), \tag{8}$$

which depends on current whale stock and total vessel capacity. Given the initial stock W_1, the stock's law of motion (1), and the aggregate harvest (8), firms form beliefs about the future path of the whale stock. For notational convenience, I combine the firm state, resource stock, and industry aggregates into a payoff-relevant state $s_{it} = (x_{it}, W_t, K_t, Q_t) \in \mathcal{S}$.

With i.i.d. entry costs, exit values, and capacity adjustment costs, incumbents follow the decision rule $\psi_t(s_{it}, \zeta_{it}, \epsilon_{it}) = a_{it}$, while potential entrants use $\psi_t(s_{it}, \kappa_{it}) = a_{it}^{\text{pe}}$. Here, the time subscript in $\psi_t(\cdot)$ indicates that decision rules are nonstationary due to the evolution of the industry, embodied in exogenous demand shifters and industry-wide technology

[22] Random shocks introduce ex ante uncertainty into firms' decisions. They help capture variability in the data that would otherwise be hard to match empirically (Rust 1987).

[23] Weintraub, Benkard, and Van Roy (2008) introduce their oblivious equilibrium as an approximation to the Markov perfect equilibrium. Their solution concept fits my framework naturally, but I assume a competitive market, not imperfect competition. In the data, the largest firm's share of the annual whale harvest averaged 3.4% over the Golden Age (1830–1860) and never exceeded 4.8%.

growth. The year-t profile of decision rules is

$$\psi_t = \{\psi_t(s_{it}, \zeta_{it}, \epsilon_{it})\}_{i \in \mathcal{I}_t} \cup \{\psi_t(s_{it}, \kappa_{it})\}_{i \in \mathcal{I}_t^{\text{pe}}}. \tag{9}$$

Let $f_t(x)$ denote the number of firms in state x in year t, and let $M_t(x' \mid x, \psi_t)$ denote the probability that a firm in state x in year t moves to state x' in year $t+1$ under the decision rules ψ_t. The distribution of firms evolves according to:

$$f_{t+1}(x') = \begin{cases} \sum_{x \in \mathcal{X}} M_t(x' \mid x, \psi_t) f_t(x) + \sum_{i \in \mathcal{I}_t^{\text{pe}}} \psi_t(s_{it}, \kappa_{it}) & \text{if} \quad x' = x^{\text{e}} \\ \sum_{x \in \mathcal{X}} M_t(x' \mid x, \psi_t) f_t(x) & \text{otherwise,} \end{cases} \tag{10}$$

where x^{e} is the state of a new entrant. The evolution of the firm distribution depends on the initial distribution f_1 and the initial resource stock W_1. For notational simplicity, I suppress the year-1 dependence in the remainder of the analysis.

3.3. Dynamic optimization

Firms make their dynamic choices of entry, exit, and capacity adjustment to maximize their expected discounted stream of profits. They discount future profits by a factor $\rho \in (0, 1)$.

A potential entrant draws an entry cost κ_{it} from an i.i.d. logistic distribution with mean $\bar{\kappa}$ and scale σ_L. This entry cost represents a barrier to profitable entry, capturing heterogeneity in firms' ability to successfully enter and operate in the industry.

An incumbent firm faces two shocks. First, it draws an exit value ζ_{it} from an i.i.d. logistic distribution with mean $\bar{\zeta}$ and the same scale σ_L, which reflects the opportunity cost of remaining in the whaling industry, including outside opportunities. Second, it draws a marginal capacity adjustment cost ϵ_{it} from an i.i.d. normal distribution with standard deviation σ.[24] Capacity adjustments involve expanding or contracting fleet size (by buying or selling vessels) and related operations. Variation in organizational frictions, financing conditions, and firm-specific constraints generates heterogeneity in adjustment costs. I assume that the three shocks (entry cost, exit value, and marginal adjustment cost) are drawn independently of one another and over time.

In addition to random shocks, incumbent firms incur deterministic adjustment costs.

[24] Stochastic adjustment costs date back at least to Blundell et al. (1992). Introducing stochasticity into *marginal* adjustment costs follows Kalouptsidi (2018), Caoui (2023), Gowrisankaran, Langer, and Reguant (2024), and Gowrisankaran and Schmidt-Dengler (2025). In contrast, Ryan (2012) and Fowlie, Reguant, and Ryan (2016) apply stochastic shocks to *fixed* adjustment costs.

These costs depend on current capacity, the magnitude of the adjustment, and whether the firm invests or divests:

$$(11)\quad C(K_{it+1}, K_{it}; \gamma) = \mathbb{1}\{K_{it+1} > K_{it}\}\left(\gamma_0^+ + \gamma_1^+ I_{it} + \gamma_2^+ I_{it}^2\right) \\ + \mathbb{1}\{K_{it+1} < K_{it}\}\left(\gamma_0^- + \gamma_1^- I_{it} + \gamma_2^- I_{it}^2\right),$$

where $\gamma = \{\gamma_0^+, \gamma_1^+, \gamma_2^+, \gamma_0^-, \gamma_1^-, \gamma_2^-\}$ and I_{it} is defined by equation (2). The first part, indicated by $\mathbb{1}\{K_{it+1} > K_{it}\}$, captures investment costs, whereas the second part, indicated by $\mathbb{1}\{K_{it+1} < K_{it}\}$, represents divestment costs. Each part comprises three elements: a fixed cost (γ_0), a linear cost (γ_1), and a convex cost (γ_2).

An active firm with state $s_{it} = (x_{it}, W_t, K_t, Q_t)$ in year t, given that all firms follow the decision rules ψ_t, solves the following dynamic problem:

$$(12)\quad V_t(s_{it}, \psi_t; \Theta) = \Pi_t(s_{it}; \alpha, \beta) + \mathbb{E}_\zeta\Bigg[\max\Bigg\{\zeta_{it}, \\ \mathbb{E}_\epsilon\left[\max_{K_{it+1}}\{-C(K_{it+1}, K_{it}; \gamma) - I_{it}\epsilon_{it} + \rho(1-p_t)\,\mathbb{E}_t\left[V_{t+1}(s_{it+1}, \psi_{t+1}; \Theta)\,|\, s_{it}, \psi_t\right]\}\,\Bigg|\, s_{it}\right]\Bigg\}\,\Bigg|\, s_{it}\Bigg],$$

where $\Theta = \{\alpha, \beta, \bar{\zeta}, \bar{\kappa}, \sigma_L, \sigma, \gamma\}$, $s_{it+1} = (x_{it+1}, \widetilde{W}_{t+1}, K_{t+1}, Q_{t+1})$, and p_t is the exogenous end-of-period wreck hazard from the timing sequence in Section 3.1.[25] Because the expectations over ζ_{it} and ϵ_{it} are taken before those shocks realize, $V_t(\cdot)$ is an ex-ante (integrated) value function. Its time subscript reflects the nonstationarity, as period profits $\Pi_t(\cdot)$ vary over time due to demand shifts and technological progress. A firm exits in year t if and only if its exit value (ζ_{it}) exceeds the expected continuation value. If it stays, the firm selects the next period's capacity level K_{it+1} to maximize its discounted expected future value net of adjustment costs. A firm that stays reaches year $t+1$ only if it survives the end-of-period wreck hazard, which occurs with probability p_t.[26] The conditional expectation $\mathbb{E}_t[\,\cdot\,|\, s_{it}, \psi_t]$ averages over the firm's productivity ω_{it+1} from (3) and uncertain stock level $\widetilde{W}_{t+1}$ from (1), while the aggregates K_{t+1} and Q_{t+1} are perceived to evolve deterministically.

[25] Note that firms form beliefs about the future whale stock $\widetilde{W}_{t+1}$ (instead of W_{t+1}), subject to uncertainty.

[26] The wreck hazard p_t is calibrated from the firm-year panel as the share of operating firms that lost their entire fleet to a wreck in year t and did not operate again. Online Appendix A.1 describes the treatment of wrecks in the moment conditions.

For a potential entrant, the objective is to solve the following problem:

$$\mathbb{E}_{\kappa}\left[\max\left\{0,\ \rho\,\mathbb{E}_t\left[V_{t+1}(s_{it+1},\psi_{t+1};\Theta)\,\middle|\,s_{it},\psi_t\right]-\kappa_{it}\right\}\,\middle|\,s_{it}\right]. \tag{13}$$

A potential entrant decides to enter at year t if and only if the discounted expected value net of the entry cost is greater than zero.[27]

3.4. Equilibrium

The model captures the rise and fall of the American whaling industry, together with shifts in technology, demand, and whale stocks over time. Because of this nonstationary nature, I solve for equilibrium via backward induction, assuming that the nonstationary equilibrium converges to a stationary equilibrium in the long run (Benkard, Jeziorski, and Weintraub 2024). Because the industry was almost non-existent by 1915, I assume that the industry reaches a stationary equilibrium at that point and solve for the nonstationary equilibrium backward from there with the terminal value function of zero.

I now define the dynamic competitive equilibrium for a common-pool resource industry. Given an initial firm distribution f_1 and resource stock W_1, an equilibrium is a sequence of decision rules $\{\psi_t\}_{\forall t}$, a price path $\{P_t\}_{\forall t}$, a stock path $\{W_t\}_{\forall t}$, and an aggregate trajectory $\{K_t, Q_t\}_{\forall t}$ such that:

1. Incumbents solve (12) and potential entrants solve (13), taking the price and the industry aggregates as given.
2. In every year, the market price is given by (5), total vessel capacity is given by (7), and total whale harvest is given by (8).
3. The firm distribution evolves according to (10).
4. The whale stock evolves according to (1).

Existence follows from the oblivious-equilibrium arguments of Weintraub, Benkard, and Van Roy (2008) and their nonstationary extension in Benkard, Jeziorski, and Weintraub (2024). In this dynamic competitive equilibrium, each firm solves a single-agent problem; the strictly convex adjustment cost pins down a unique capacity choice, and the continuous shocks smooth the exit and adjustment margins.[28]

[27] Equations (12) and (13) normalize the operating fixed cost to zero. This normalization is necessary because, in a dynamic setting, the entry cost, exit value, and operating fixed cost cannot be jointly identified (Aguirregabiria and Suzuki 2014). As a result, estimates of entry cost and exit value incorporate the operating fixed cost.

[28] I test uniqueness numerically, solving from a wide range of initial guesses and always recovering the same fixed point.

TABLE 2. List of parameters and empirical approach

Parameters	Notation	Empirical approach
Panel A. Period payoffs		
Harvesting function	$\beta_0, \beta_k, \beta^K, \beta^W, \beta^t, \lambda$	DP-GMM (Section 4.1)
Demand curve	$\alpha_0, \alpha_p, \alpha^{\text{gdp}}, \alpha^t, \alpha^{\text{pet}}$	OLS, IV (Section 4.1)
Panel B. Dynamic costs and values		
Mean of entry costs	$\bar{\kappa}$	NFXP-GMM (Section 4.2)
Mean of exit values	$\bar{\zeta}$	NFXP-GMM (Section 4.2)
Scale of entry costs and exit values	σ_L	NFXP-GMM (Section 4.2)
SD of stochastic marginal adjustment costs	σ	NFXP-GMM (Section 4.2)
Deterministic adjustment costs	$\gamma_0^+, \gamma_1^+, \gamma_2^+, \gamma_0^-, \gamma_1^-, \gamma_2^-$	NFXP-GMM (Section 4.2)
Panel C. Transitions		
Annual discount factor	$\rho = 0.95$	Calibrated
Vessel capacity depreciation rate	$\delta = 0.05$	Calibrated
Whale stock dynamics	$z = 1.4, r = 0.011$	Calibrated
Belief dispersion over the whale stock	$\sigma_W = 0.10$	Calibrated
Destruction hazard	p_t	Calibrated

Note: This table summarizes key parameters of the model. Parameters in panels A and B are estimated, whereas those in Panel C are calibrated. In panel A, DP-GMM refers to dynamic panel estimation using the generalized method of moments. In panel B, NFXP-GMM denotes nested fixed-point estimation by the generalized method of moments. Parameters related to whale population dynamics are drawn from previous literature (Baker and Clapham 2004; Whitehead 2002). The preferred harvesting-function estimates, column (4) of Table 3, exclude crew size, so β_l does not enter the estimated model.

4. Estimation

The estimation proceeds in two stages. In the first stage, I estimate the whale harvesting (production) function and the whale demand curve (Section 4.1). These estimates provide period payoffs for all possible firm states and years. In the second stage, I embed these payoffs into the dynamic model and estimate the dynamic parameters (Section 4.2). Table 2 summarizes the structural parameters that are estimated and calibrated. Panel A reports the primitives from the harvesting function and demand curve. Panel B lists the dynamic parameters. Panel C includes calibrated values.

4.1. Period payoffs

Whale harvesting function. I parameterize equation (4) as a Cobb–Douglas production function in logarithmic form:

$$q_{it} = \beta_0 + \beta_k k_{it} + \beta_l l_{it} + \beta^K k_t + \beta^W w_t + \beta^t t + \underbrace{\mu_{it} + \varepsilon_{it}}_{\omega_{it}}, \tag{14}$$

where q_{it} represents the log of the number of harvested whales (sperm whales + baleen whales), k_{it} is the log of vessel capacity, l_{it} is the log of crew size, k_t is the log of aggregate vessel capacity, w_t is the log of whale stock, and t is the time trend. The firm-level (log) productivity, ω_{it}, is divided into persistent productivity μ_{it} and transitory shock ε_{it}.

Following the dynamic panel approach by Blundell and Bond (2000), persistent productivity μ_{it} evolves as an AR(1) with serial correlation λ:

$$\mu_{it} = \lambda\mu_{it-1} + \xi_{it}, \tag{15}$$

where ξ_{it} is an independent and identically distributed (i.i.d.) shock. The split between μ_{it} and ε_{it} matters for identification.[29]

Firms choose vessel capacity in year $t-1$ (as in Eq. 2) and crew size between $t-1$ and t (as in Ackerberg, Caves, and Frazer 2015), both prior to the year t productivity shock ξ_{it}. Combined with the AR(1) process (Eq. 15), this timing yields the exclusion restriction identifying $\beta = (\beta_0, \beta_l, \beta_k, \beta^K, \beta^W, \beta^t, \lambda)$. Specifically, the year-t productivity shock ξ_{it} is orthogonal to $z_{it} = (1, l_{it-1}, k_{it}, k_{it-1}, k_t, w_t, t)$. The corresponding moment conditions are

$$\mathbb{E}\left[\widehat{\xi_{it} + (\varepsilon_{it} - \lambda\varepsilon_{it-1})}(\beta) \otimes z'_{it}\right] = 0.$$

I estimate the parameters by nonlinear generalized method of moments (GMM). The moment conditions are exactly identified, so I solve them directly rather than weighting them.

Table 3 reports estimates of the whale harvesting function. Columns (1)–(3) use the subsample that contains crew data, while column (4) uses the full sample and serves as the baseline for the dynamic analysis. Column (1) only includes vessel capacity and crew size; column (2) adds aggregate time series variables, which absorb much of the serial correlation and reduce the estimated persistence parameter. Column (3) excludes crew size from the same subsample to compare with the full-sample specification in column (4).

The estimated results show that vessel capacity alone, without crew size, explains most

[29] Imposing this AR(1) structure rules out the richer productivity transition in equation (3) and may not fully address selection bias from endogenous exit. Fully resolving the selection issue would require estimating the production function jointly with firms' entry, exit, and capacity adjustment decisions, which could be an extension I leave for future work. A common alternative is an approach that inverts intermediate-input demand to recover latent productivity (Levinsohn and Petrin 2003; Ackerberg, Caves, and Frazer 2015). This approach requires an intermediate input whose demand is monotonically increasing in productivity. Because the data for the American whaling industry lack such an input, this approach is not feasible here. Instead, in Online Appendix Table F6, I test the dynamic estimates by increasing $\hat{\beta}_k$, because the endogenous exit concerns downward bias of $\hat{\beta}_k$. The results are not sensitive to this change.

TABLE 3. Estimates of the whale harvesting function

	(1)	(2)	(3)	(4)
Vessel capacity: β_k	1.010	0.872	1.060	1.052
	(0.028)	(0.027)	(0.009)	(0.007)
Crew size: β_l	0.003	0.235		
	(0.025)	(0.030)		
Persistence: λ	0.634	0.227	0.321	0.248
	(0.107)	(0.110)	(0.112)	(0.088)
Whale stock: β^W		6.121	5.747	4.899
		(0.282)	(0.301)	(0.245)
Aggregate vessel capacity: β^K		-0.041	-0.026	-0.056
		(0.015)	(0.016)	(0.009)
Time trend: β^t		0.017	0.017	0.013
		(0.001)	(0.002)	(0.001)
Returns to scale: $\beta_k + \beta_l$	1.013	1.106	1.060	1.052
Number of firms	528	528	528	1041
Observations	5,346	5,346	5,346	9,601

Notes: This table reports the estimated harvesting function. Columns (1)–(3) use the subsample with crew data; column (4) employs the full sample. Identification relies on the following instruments: contemporaneous and lagged vessel capacity, lagged crew size, contemporaneous aggregate fleet capacity, whale-stock abundance, and a linear time trend. Parameters are estimated by generalized method of moments (GMM). Robust GMM standard errors, based on the estimator's asymptotic variance–covariance matrix, appear in parentheses.

of the variation in production scale. This pattern is intuitive because nineteenth-century American whaling was capital-intensive, and crew size was roughly proportional to vessel capacity. Accordingly, in the analyses that follow, I assume that harvesting combines vessel capacity and crew in fixed proportions, reflecting a Leontief technology.[30]

Based on the results of column (4), a back-of-the-envelope calculation suggests strong production externalities. First, the coefficient for whale stock (β^W) captures the stock externality. Holding other factors constant, a 1 percent increase in whale stock raises an individual firm's harvest by 4.9 percent. Because whale stock fell by 15.4 percent between 1820 and 1850, the implied harvest reduction is 56.0 percent. Second, the coefficient for aggregate capacity (β^K) captures the congestion externality. A 1 percent increase in aggregate capacity reduces an individual firm's harvest by 0.056 percent. Between 1820 and 1850, aggregate capacity rose from 39,242 to 185,390 tons (an increase of 372 percent). Applying the elasticity estimate implies an 8.3 percent decline in an individual firm's harvest due to

[30]I therefore model crew size as $L(K_{it}) = cK_{it}$, where c denotes crew per tonnage. The data support this assumption: vessel tonnage and crew size are tightly linked. The correlation coefficient between capacity K_{it} and crew size L_{it} is 0.92; between their logs k_{it} and l_{it} it is 0.9.

congestion.

Demand for whales. The whaling market features a demand curve in the form of equation (5), inverted and parameterized as:

$$\ln Q_t = \alpha_0 + \alpha_p \ln P_t + \alpha^{\mathrm{gdp}} \ln \mathrm{GDP}_t + \alpha^t t + \alpha^{\mathrm{pet}} \ln \mathrm{Pet}_t + \eta_t, \tag{16}$$

where P_t is the whale price, GDP_t denotes U.S. real GDP per capita, t is a time trend, and Pet_t, included for $t \geq 1859$, represents the price of petroleum. η_t is an i.i.d. demand shock. I estimate this specification separately before and after 1859, as demand likely shifted after the discovery of petroleum (see Section 2.1 for details).

The whale price may be endogenous because it can co-move with the demand shock η_t. Such shocks shift the demand curve and the equilibrium price, biasing the estimated price elasticity. I address this concern with two supply-side cost shifters as instruments for price. They move supply costs and hence prices but are orthogonal to η_t conditional on controls.

First, I use whale stocks lagged five years. Because whales are slow-growing, past abundance strongly predicts current stock levels, which affect harvest costs. A five-year lag also exceeds the typical voyage length of two to three years, reducing simultaneity with current harvest outcomes. Current demand for whale products cannot affect whale abundance five years earlier. Any reverse channel would require persistent unobserved shocks that link past harvesting effort to current demand. Macroeconomic controls (GDP per capita and a time trend) absorb the main aggregate demand drivers, so the remaining correlation would have to come from unobserved demand shocks with unusually long persistence that both influenced harvest effort five years earlier and predict current U.S. demand, which seems implausible.

Second, I exploit variation in long-term bond yields in New England.[31] Whaling was highly capital-intensive; firms relied on external finance to purchase ships, outfit crews, and finance long voyages (see Section 2.2). Higher bond yields increase borrowing costs and therefore raise the cost of supply. Whale products, by contrast, were sold nationally and internationally, so the marginal buyer was a household or mill far from New England. After controlling for aggregate U.S. demand conditions (GDP per capita and a time trend), regional long-term borrowing costs should be orthogonal to shocks in whale-product demand. Local credit conditions largely reflected banking cycles, lending practices, and

[31] The data are constructed by Homer and Sylla (1996).

TABLE 4. Estimates of the whale demand curve

	OLS		IV	
	(1)	(2)	(3)	(4)
Whale price: α_p	-2.025	-0.211	-4.457	-4.457
	(0.978)	(0.073)	(0.793)	
US GDP per capita	-4.364	0.463	3.160	2.229
	(4.583)	(0.265)	(2.564)	(1.773)
Time trend	0.170	-0.048	0.138	-0.114
	(0.035)	(0.006)	(0.025)	(0.034)
Petroleum price		0.055		-0.363
		(0.035)		(0.361)
Observations	55	51	51	51
Sample period	1804-1858	1859-1909	1804-1858	1859-1909
First-stage F-statistic			38.392	
Overid. test (p-value)			0.25	
Adjusted R-squared	0.821	0.956	0.769	0.507

Notes: This table reports estimates of the whale demand curve. Columns (1) and (2) present OLS results for the pre- and post-petroleum periods, respectively. Column (3) reports instrumental-variables estimates for the pre-petroleum sample, instrumenting the whale price with two supply-side shifters: the whale stock lagged five years and long-term New England bond yields. Column (4) presents post-petroleum estimates, fixing the price coefficient, $\hat{\alpha}_p$, at the elasticity from column (3). The first-stage F-statistic is the robust Wald statistic for the excluded instruments; the overidentification test reports the p-value of Hansen's J statistic for their joint validity, where a large p-value fails to reject. Standard errors are reported in parentheses (Newey–West for the OLS columns, heteroskedasticity-robust for the GMM column).

regional capital scarcity rather than demand for whale products (Lamoreaux 1994).

I restrict the IV strategy to the pre-petroleum era (before 1859) for two reasons. First, whale stock exhibits little variation after 1859, undermining its validity as an instrument (Panel B of Figure 1). Between 1859 and 1909, whale stocks remained nearly constant (1.15–1.21 million whales), in contrast to the sharp decline from 1.48 to 1.18 million by 1858. Second, by the mid-century the whaling center shifted from New Bedford (and New England more broadly) to San Francisco, making New England bond yields no longer a credible cost shifter. Accordingly, for the post-petroleum period, I calibrate the price coefficient α_p using the elasticity estimated from the pre-petroleum period. I also consider a reasonable range of alternative price elasticities, re-estimate the dynamic model, and recompute the counterfactuals. The main welfare and policy implications are not sensitive to these assumptions.

Table 4 reports estimates of the whale demand curve. Columns (1) and (2) present OLS estimates for the pre- and post-petroleum eras, with price elasticities of −2.025 and −0.211, respectively. Column (3) shows IV estimates for the pre-petroleum sample; the elasticity rises in absolute magnitude to −4.457, consistent with Kaiser (2013). For the post-petroleum period, I fix the price coefficient α_p at −4.457 and estimate the remaining parameters via OLS. The results in columns (3) and (4) form the basis for subsequent dynamic analyses. Online Appendix Table F5 repeats the dynamic estimation and the counterfactuals of Sections 4.2 and 5 with the post-petroleum price coefficient fixed at −2.0 and at −5.0.

4.2. Dynamic costs and values of whaling

I estimate the dynamic parameters by nested fixed-point generalized method of moments (GMM). Online Appendix D discusses the choice of a full-solution method. For each candidate parameter vector, I solve the equilibrium, which gives the probabilities of exit, of each capacity level, and of entry at every state and year.

The main computational challenge comes from continuous capacity adjustment. Computing optimal decision rules for dynamic actions can be highly sensitive to how the continuous state (and action) space is discretized. To address this issue, I adopt the approach of Gowrisankaran and Schmidt-Dengler (2025), which introduces random shocks to marginal adjustment costs and yields choice probabilities that are insensitive to the discretization. Under this approach a capacity level that no draw of the adjustment-cost shock makes optimal (dominated action) has probability zero, which leaves the log likelihood undefined, so I estimate by GMM, as in Gowrisankaran, Langer, and Reguant (2024). Online Appendix A.1 derives the choice probabilities and defines the moments.

The moments compare firms' observed decisions with the model's predictions at each firm's state and year. On the exit margin, the moment is the difference between the data and the model in the exit rate. On the capacity margin, among staying firms, the moments are the differences in the shares of firms that invest and that divest, in the amounts invested and divested, and in the shares of firms that move at least two grid steps in either direction. On the entry margin, the moment is the difference between the number of entrants in a year and the number the model predicts from the pool of potential entrants. I interact the exit moments and the shares of firms that invest and divest with the firm's capacity and productivity, the amounts invested and divested with capacity, and the exit and entry moments with the year. Online Appendix A.1 defines the 18 moments.[32]

[32]Identification comes from firms' observed decisions to enter, adjust capacity, and exit, given expected future profits across states and years within each phase. The level of exit identifies the mean exit value $\bar{\zeta}$. How exit responds to capacity, productivity, and the year identifies the scale σ_L of the entry cost and exit value distributions, because these variables move the continuation value and σ_L sets how strongly the exit probability responds to it. Given σ_L, the level of entry identifies the mean entry cost $\bar{\kappa}$. The frequencies of investing and divesting identify the fixed costs γ_0^+ and γ_0^-. Interacting the investment and divestment indicators with capacity separates the fixed costs from the per-ton terms. The fixed cost of a move is the

TABLE 5. Estimates of whaling industry dynamics

	Unit	1815-29	1830-44	1845-59	1860-74	1875-89	1890-1905
Panel A. GMM estimates							
Mean of entry costs: $\bar{\kappa}$	\$000	315.94	390.23	874.44	1078.08	1168.61	1038.46
		(38.53)	(22.54)	(60.33)	(235.86)	(137.21)	(213.58)
Mean of exit scrap values: $\bar{\zeta}$	\$000	188.42	267.45	671.82	868.76	964.79	715.55
		(37.59)	(25.22)	(51.39)	(204.92)	(129.08)	(151.63)
Scale parameter: σ_L	\$000	73.53	64.27	89.81	82.06	116.26	143.91
		(17.99)	(9.54)	(10.00)	(14.33)	(30.61)	(52.24)
Investment fixed: γ_0^+	\$000	5.92	8.84	11.04	5.89	0.00	23.91
		(3.47)	(1.58)	(1.77)	(3.45)	(9.28)	(20.23)
Investment linear: γ_1^+	\$/ton	343.37	284.07	308.50	461.89	858.99	1000.00
		(32.90)	(17.67)	(16.13)	(92.55)	(148.53)	(204.78)
Investment convex: γ_2^+	\$/ton^2	0.33	0.25	0.29	0.33	0.65	1.00
		(0.06)	(0.03)	(0.03)	(0.11)	(0.16)	(0.26)
Divestment fixed: γ_0^-	\$000	13.28	22.80	20.79	28.65	68.61	68.91
		(5.93)	(3.41)	(2.61)	(10.32)	(19.10)	(23.53)
Divestment linear: γ_1^-	\$/ton	0.00	0.00	65.92	121.18	0.00	0.00
		(63.02)	(27.45)	(20.49)	(41.92)	(123.67)	(171.45)
Divestment convex: γ_2^-	\$/ton^2	0.33	0.31	0.26	0.40	1.10	0.64
		(0.09)	(0.04)	(0.03)	(0.14)	(0.28)	(0.19)
SD of marginal adj. costs: σ	\$/ton	224.94	221.38	206.06	299.18	736.78	749.21
		(43.21)	(25.53)	(22.28)	(103.17)	(160.52)	(157.67)
Number of observations		1097	3266	3511	1690	751	371
Panel B. Realized magnitudes							
Entry cost	\$000	218	303	710	911	993	745
Exit value	\$000	409	478	945	1102	1304	1102
Investment of 300 tons	\$000	38	34	44	35	7	39
Divestment of 300 tons	\$000	67	44	47	55	68	125

Note: This table reports estimates of dynamic parameters for entry, exit, and capacity adjustment. All dollar values are adjusted to 1880 prices. Each column presents results for the corresponding period. Panel A summarizes two-step GMM estimates from the baseline set of 18 moments (Online Appendix A.1). Standard errors, shown in parentheses, are computed from the GMM sandwich formula with the covariance of the moments clustered by firm. Panel B reports the average cost or value realized by the firms that took each action, over 200 simulations of the estimated model. Investment and divestment are for a change of 300 tons, close to the median adjustment size in Table 1, and divestment is the amount the firm receives.

I estimate the model in six phases: 1815–1829, 1830–1844, 1845–1859, 1860–1874, 1875–1889, and 1890–1905. I divide the sample this way because the industry faced structural breaks that plausibly shifted firms' beliefs (Section 2.1). The arrival of petroleum and the Civil War make 1860 a natural breakpoint, and the mid-1840s shocks (the California gold rush, longer voyage durations, etc.) motivate a separate phase beginning in 1845. I divide the remaining phases into 15-year intervals for consistency. Finally, because the War of 1812 (1812–1815) severely disrupted the industry and left little firm-level data, the estimation sample begins in 1815.

Table 5 reports the dynamic parameter estimates. Panel A shows that the estimates are economically plausible and precisely estimated. Mean entry costs range from $315,940 to $1,168,610 across the six phases, low during the industry's expansion through 1844 and two to three times higher from 1845 on. Mean exit scrap values are lower in every phase, ranging from $188,420 to $964,790, and rise on the same schedule. Low entry costs and low exit values during the expansion are what the observed turnover requires, with heavy entry and limited exit while the industry grew. Adjustment cost estimates indicate sizable frictions in changing capacity. The linear cost terms imply that investment costs rise with scale, ranging from $284.07 to $1,000.00 per ton, while divestment values range from $0 to $121.18 per ton.[33] Convex cost parameters range from $0.25 to $1.00 per ton squared for investment and from $0.26 to $1.10 for divestment. Finally, the standard deviation of marginal adjustment-cost shocks, ranging from $206.06 to $749.21 per ton, indicates substantial heterogeneity in adjustment intensity.

Panel B of Table 5 reports the costs and values that firms realized, averaged over the firms that took each action in 200 simulations of the estimated model. Firms enter on low cost draws, leave on high exit values, and adjust on favorable shocks, so the realized magnitudes differ from the (unconditional) means in Panel A. Adding 300 tons of capacity cost $44,000 in 1845–1859. According to the historical record, buying a 300-ton ship and outfitting a 39- to 46-month voyage cost $41,000 to $46,000 in the same years (Davis, Gallman, and Gleiter 2007). Divesting 300 tons returned $44,000 to $125,000 across the phases. Realized entry costs and exit values are an order of magnitude larger, $218,000 to $993,000 and $409,000 to $1,304,000 because the entry cost includes the initial business setup cost

same at every size. The mean amounts invested and divested, and how they scale with capacity, identify the linear and convex terms γ_1^+, γ_2^+, γ_1^-, and γ_2^-, since a higher convex term makes large one-time adjustments more costly. The frequencies of moves of two grid steps or more, together with the convex terms, identify the standard deviation σ of the marginal adjustment cost shock, because a larger σ spreads the chosen capacity levels further from the level a firm would choose without the adjustment cost shock.

[33]When firms divest, a positive γ_1^- implies a positive linear "value." Total adjustment costs should be assessed jointly with fixed and convex costs; see Panel B of Table 5.

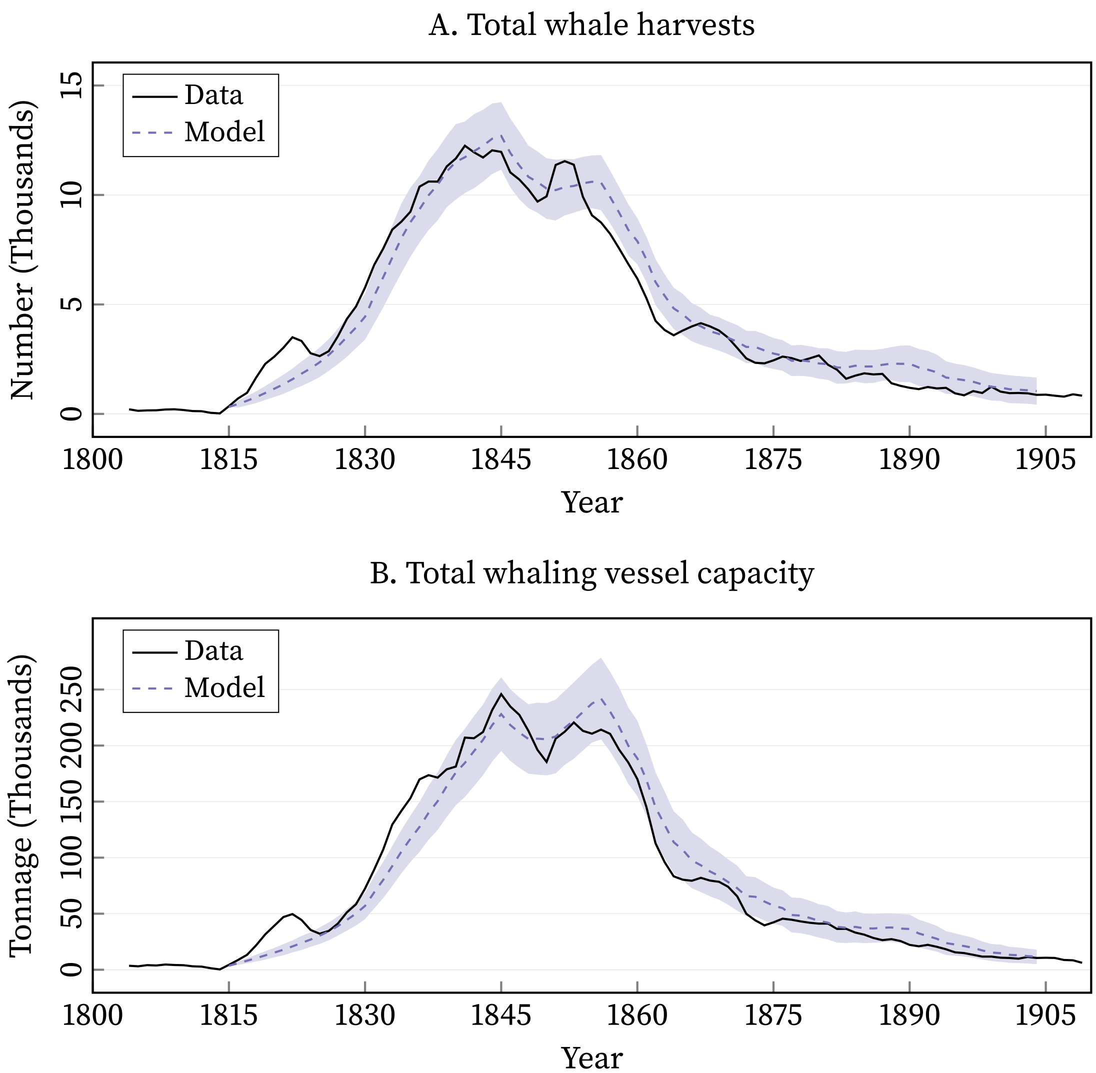

FIGURE 3. Fit of the estimated model

Note: Panel A shows total whale harvest; Panel B shows total whaling-vessel capacity. The black solid line indicates the actual data, and the purple dashed line shows the model-predicted paths. Model outcomes are averaged over 200 simulations of the estimated model, with shaded bands denoting 95 percent confidence intervals.

and all kinds of the barrier to becoming a whaling firm, and the exit value captures the firm's next-best use of the capital (opportunity cost of staying in the whaling industry).

Figure 3 compares the simulated paths of total whale harvest and total vessel capacity with the data. The estimated model reproduces the industry's rise and fall over nine decades. Online Appendix E discusses the fit in detail.

5. Counterfactuals

Based on the estimated model, this section examines the social cost of externalities in the long-run dynamics. Section 5.1 states the planner's problem to recover the shadow value of a whale. Section 5.2 reports the welfare loss under open access in comparison to the social planner. Section 5.3 asks how the demand for whale products and the industry's technology impact the scale of welfare loss.

5.1. The planner's problem and the shadow value of a whale

I define the social welfare function in year t as the sum of consumer and producer surplus:

$$\begin{aligned}\mathbb{W}_t = &\underbrace{\int_0^{Q_t} \mathcal{P}_t(\varphi)\mathrm{d}\varphi - \mathcal{P}_t(Q_t)Q_t}_{\text{consumer surplus}} \\ &\underbrace{+ \sum_{i\in\mathcal{I}_t} \left\{\Pi_t(s_{it}) + \mathbf{1}_{\{a_{it}=0\}}\zeta_{it} - \mathbf{1}_{\{a_{it}>0\}}\left[C(I_{it}) + I_{it}\epsilon_{it}\right]\right\} - \sum_{i\in\mathcal{I}_t^{\mathrm{pe}}} \mathbf{1}_{\{a_{it}^{\mathrm{pe}}=1\}}\kappa_{it}}_{\text{producer surplus}},\end{aligned} \tag{17}$$

A social planner chooses incumbents' exit and capacity adjustments $\{a_{it}\}_{\forall i,t}$ together with potential entrants' entry decisions $\{a_{it}^{\mathrm{pe}}\}_{\forall i,t}$ to maximize the discounted sum of social welfare (17) subject to the whale stock's law of motion (1). The Lagrangian is

$$\mathbb{E}\left[\sum_t \rho^{t-1}\left\{\mathbb{W}_t + \rho\Lambda_{t+1}^W\Big(\Big(W_t + rW_t\big[1-(W_t/W_1)^z\big] - Q_t\Big)\phi_{t+1}^W - W_{t+1}\Big)\right\}\right], \tag{18}$$

where Λ_{t+1}^W refers to the multiplier to the whale-stock constraint. The expectation is taken with respect to the cost and value shocks $(\zeta_{it}, \epsilon_{it}, \kappa_{it})$, the productivity draws (Ω_{it}) and the stock shock ϕ_{t+1}^W.

The shadow value of a whale. The multiplier Λ_{t+1}^W is the shadow value of a whale in the year-$(t+1)$ stock, measured in year-$(t+1)$ dollars. A whale harvested in year t is a whale removed from that stock, so what it costs society, in year-t dollars, is $\rho\Lambda_{t+1}^W$.[34] Intuitively, Λ_{t+1}^W is the discounted value of every future harvest that one more whale in the stock would have yielded. What that whale is worth therefore depends on the firms and vessels that will be there to take it, and those are themselves the outcome of past entry, exit, and capacity

[34] Online Appendix A.2 derives the shadow value from the planner's first-order condition.

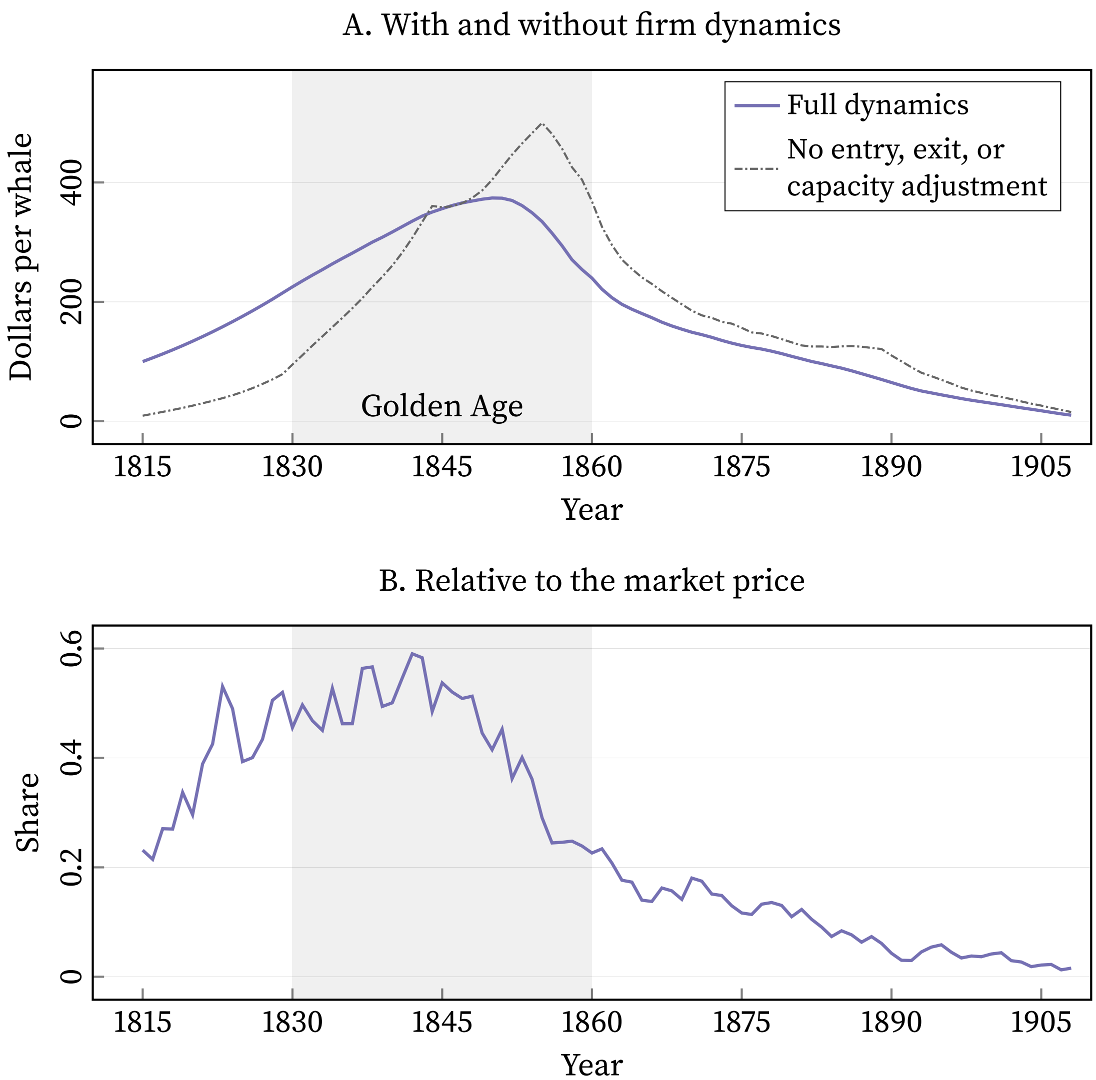

FIGURE 4. The shadow value of a whale

Note: In panel A, the solid line propagates a marginal whale through the estimated equilibrium with entry, exit, and capacity adjustment all active. The dash-dotted line removes all three dynamics. Panel B divides the solid line of panel A by the observed market price of a whale in the same year. The shaded band marks the Golden Age (1830–1860).

adjustment.

Figure 4 shows the measured shadow value. In panel A, the solid line is the shadow value under the full dynamic model; the dash-dotted line is the shadow value without firm dynamics (i.e., the fleet is held fixed). A whale landed in 1815 has the true social value of \$100 under the full dynamic model and \$9 with the fleet held fixed. Nine-tenths of what a whale was worth in 1815 was therefore driven by the fleet that had not yet been built. Over the early boom (1815–1830), that fleet accounts for two-thirds of the external cost of the whales harvested. Around 1845 the ordering turns over. By 1855 the full shadow value is

TABLE 6. The welfare cost of open access

	Laissez-faire	First best	Difference
Social welfare	46.37	60.65	14.28
Share of first best (%)			[23.5]
By era			
Pre-1859	30.73	52.53	21.80
Post-1859	15.64	8.12	−7.52
By welfare component			
Consumer surplus	13.89	8.06	−5.83
Producer surplus	32.48	52.58	20.11

Note: All entries are in millions of 1880 dollars, discounted to 1815 at $\rho = 0.95$ and averaged over 300 simulations. Laissez-faire is the estimated equilibrium; the first best solves problem (18). The era rows split each allocation's welfare at the petroleum discovery, and the component rows split it into consumer and producer surplus.

$335 while the fixed-fleet shadow value is $500, because the fleet that would take that whale is by then shrinking rather than growing. In sum, overlooking entry, exit, and capacity adjustment understates the shadow value of whales in the early boom and overstates it in the late bust.

Panel B divides the whale shadow value under the full dynamic model by what a whale sold for in the same year. The ratio is the external cost of a whale as a share of its market price. The share rose from 20 percent in 1815 to near one-half by the late 1820s and stayed there through 1850, peaking at 60 percent in the early 1840s. It fell to 25 percent by the late 1850s, as the price of a whale rose by about 80 percent between 1848 and 1856 and the shadow value began to fall, and then to 12 percent by 1875 and 4 percent by 1890. After the petroleum discovery the fleet shrank, so fewer vessels were there to take the harvests that one more whale in the stock would have yielded.

5.2. The cost of open access

Table 6 summarizes welfare outcomes under open access (laissez-faire) and the first best (social planner). Laissez-faire delivers $46.37 million and the first best $60.65 million. Open access cost $14.28 million, 23.5 percent of first-best welfare.

The gain is concentrated in time. Before 1859 the first best is worth $21.80 million more than laissez-faire, as most of the externalities were generated during the early years. By welfare component, consumer surplus falls from $13.89 million to $8.06 million, because the planner harvests fewer whales and the price of a whale is higher. Producer surplus rises from $32.48 million to $52.58 million, entirely driving the social welfare gain.

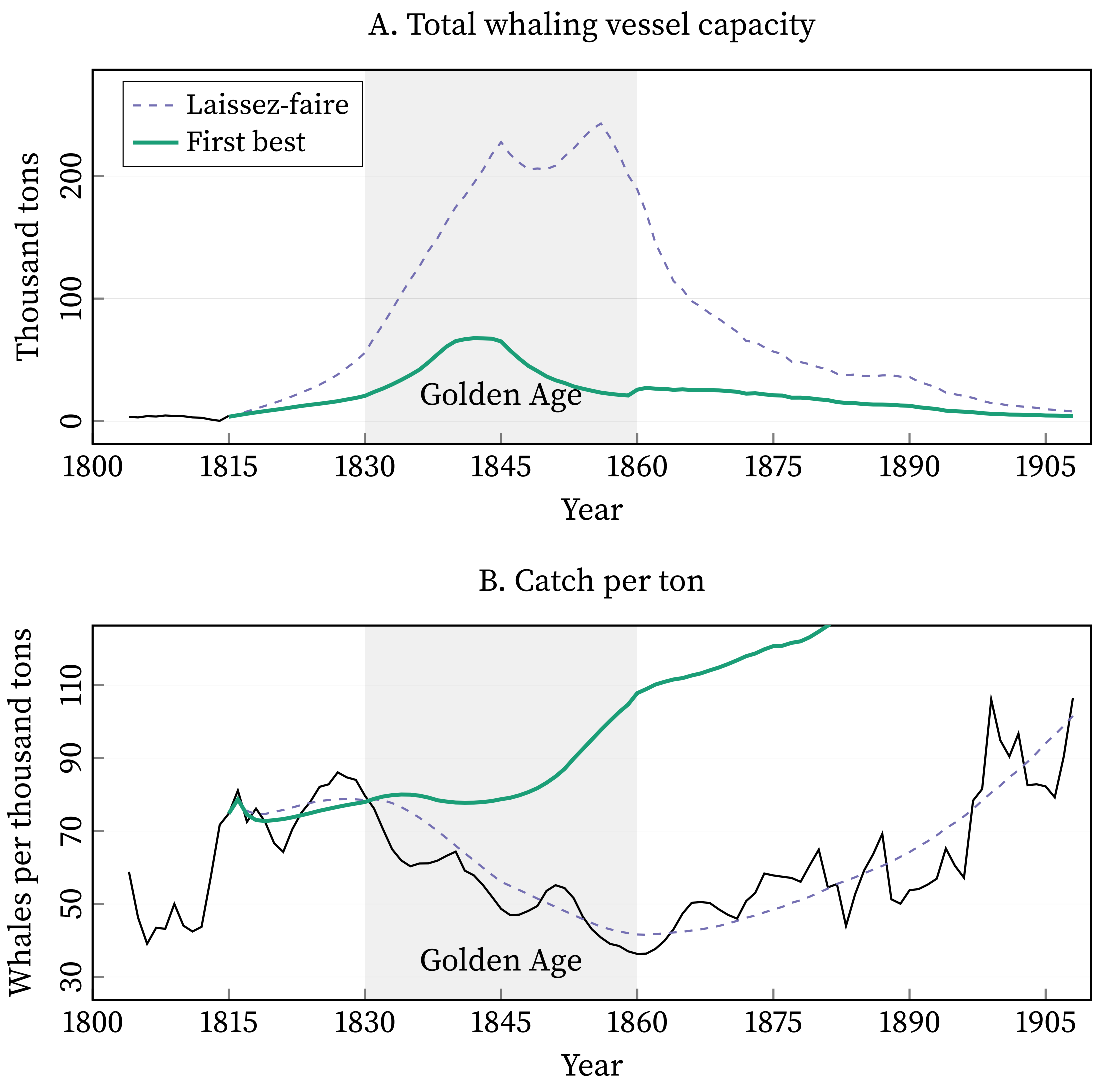

FIGURE 5. Vessel capacity and harvesting efficiency

Note: Panel A shows total whaling vessel capacity in thousand tons. Panel B shows whales harvested per thousand tons of vessel capacity, and the black line is the observed series over the whole sample. In both panels the purple dashed line is the laissez-faire equilibrium and the green solid line is the first best. The shaded band marks the Golden Age (1830–1860).

Figure 5 shows the mechanism of producer surplus gain. Panel A plots total vessel capacity under laissez-faire and under the first best. Laissez-faire vessel capacity peaks at 242.9 thousand tons in the mid-1850s, 3.6 times the first-best peak of 67.7 thousand tons. Panel B shows the efficiency consequence. It plots the number of whales harvested per thousand tons of vessel capacity, in the data, under laissez-faire, and under the first best. Under laissez-faire, harvesting efficiency falls from 78.5 whales per thousand tons in 1815–1819 to 43.2 in 1855–1859, a decline of 45 percent. The first best moves the other way. It rises by 31 percent over the same window. By the late 1850s a ton of first-best capacity harvests

TABLE 7. The cost of open access by petroleum arrival date

	Petroleum arrival year			
	1859	1875	1890	Never
Panel A. Social welfare				
Laissez-faire ($M)	46.37	38.97	29.76	20.36
First best ($M)	60.65	63.12	69.15	78.05
Difference ($M)	14.28	24.14	39.39	57.68
Share of first best (%)	[23.5]	[38.3]	[57.0]	[73.9]
Panel B. Laissez-faire outcomes				
Lowest stock level (Million whales)	1.14	1.07	0.98	0.85
Peak capacity (Thousand tons)	242.9	260.2	364.4	484.0

Note: Millions of 1880 dollars, 300 paired draws. Each column re-solves laissez-faire and the first best with the petroleum discovery moved to the stated year; the pool of potential entrants is held at its historical path and the post-petroleum demand curve is shifted so that it begins in the counterfactual arrival year. In the Never column the pre-petroleum demand curve applies through the end of the horizon. The 1859 column is the baseline of Table 6. The stock trough and the peak capacity are laissez-faire outcomes.

2.3 times what a ton under laissez-faire harvests. Under open access, every additional ton draws down the whale stock and crowds the hunting grounds, so every ton harvests less compared to the first best.

5.3. Whaling's fortunes and externalities

The welfare loss in Table 6 is the cost under the demand and technology paths the industry actually faced. Demand for whale products grew with the American economy until the petroleum discovery of 1859 reduced it, and harvesting technology gradually improved throughout the century. This subsection re-solves both allocations under counterfactual paths for each.

The petroleum discovery. Petroleum was discovered at Titusville in 1859, and whale-product demand fell away over the following decade. The demand shock substantially impacted the industry before the whale stock ran out. Table 7 asks what open access would have cost had the substitute arrived later or never, re-solving both the model and the first best with the discovery moved to 1875, to 1890, and removed altogether.

The commons loss rises with each later arrival date, from $14.28 million to $24.14 million and $39.39 million, and to $57.68 million with no petroleum discovery at all, from 23.5 to 73.9 percent of first-best welfare. The arrival of petroleum in 1859 therefore reduced the cost of open access to a quarter of what it would have been. The laissez-faire industry builds the capital that does the damage. Peak vessel capacity rises from 242.9 thousand

TABLE 8. The cost of open access by technology growth rate

	Technology growth (per year)		
	0%	1.3% (estimated)	2.7%
Panel A. Social welfare			
Laissez-faire ($M)	48.46	46.37	44.69
First best ($M)	54.79	60.65	67.63
Difference ($M)	6.33	14.28	22.94
Share of first best (%)	[11.6]	[23.5]	[33.9]
Panel B. Laissez-faire outcomes			
Lowest stock level (Million whales)	1.27	1.14	1.01
Peak capacity (Thousand tons)	201.3	242.9	254.8

Note: Millions of 1880 dollars, 300 paired draws. Each column re-solves laissez-faire and the first best with the harvesting-function trend set to the stated growth rate; the 1815 level of technology is held fixed, so the columns differ only in how fast technology improves from there. The 1.3 percent column is the baseline of Table 6, at the estimated trend; the 2.7 percent column doubles the estimated slope. The stock trough and the peak capacity are laissez-faire outcomes.

tons under the actual arrival to 484.0 thousand with no arrival, and the lowest level of whale stock falls to 0.85 million whales against the 1.14 million actually reached.

First-best welfare rises with each later arrival date, from $60.65 million to $78.05 million, because demand that lasts longer is worth more under management. Laissez-faire welfare falls, from $46.37 million to $20.36 million, because the unregulated industry responds to a longer whale-products era with more vessels on a smaller whale stock. As a result, what looked disastrous of petroleum discovery was actually an event that alleviated the tragedy of the commons.

Technology growth. The harvesting (production) function estimates imply overall productivity growth of about 1.3 percent per year (Table 3). The growth reflects better gear, larger and faster vessels, and accumulating knowledge of the hunting grounds, from the toggle iron of 1848 to bomb lances in the 1850s and steam-powered whalers by the 1880s (Davis, Gallman, and Gleiter 2007). Table 8 re-solves both the model and the first best with the technological trend removed and with its slope doubled.

Technology growth of 1.3% (estimated) more than doubled the cost of open access, compared with the no advancement case. If there were no such growth, the welfare loss would have been $6.33 million, 11.6 percent of first-best welfare; at the estimated 1.3 percent per year it is $14.28 million; at double the estimated rate it is $22.94 million, 33.9 percent of the first best. Peak vessel capacity rises from 201.3 to 254.8 thousand tons across the columns, and the whale stock's lowest level falls from 1.27 to 1.01 million whales.

As in the petroleum exercise, first-best welfare rises with the technology growth rate, from \$54.79 million to \$67.63 million, while laissez-faire welfare falls, from \$48.46 million to \$44.69 million. The unregulated industry converts better harvesting technology into more capacity and a deeper decline of the whale stock, and ends up with lower welfare than it would have had without the improvement. What drove the industry's fortunes also exacerbated the tragedy of the commons.

6. Conclusion

This paper studies the commons problem through a model of firm dynamics. Using a newly constructed firm-level panel of the American whaling industry, I estimate this model and conduct counterfactual analyses. The framework in this paper allows for empirical evaluations of the common-pool resource problem in the long run, which should naturally incorporate firms' entry, exit, and investment decisions in contrast to short-run cases.

Two lessons extend beyond whaling. First, valuing a shared resource requires considering firms' dynamic actions. The shadow value of a resource depends on the capacity that will be there to take it, so ignoring the future industry evolution could misprice the resource. Second, firms' responses to demand and technology determine the scale of the tragedy. Demand that lasts longer and technology that improves faster both raise the private return to resource exploitation, and the unregulated industry converts the higher return into unpriced entry and investment, so the welfare loss grows with the industry's prospects.

Several extensions remain for future work. One is to allow spatially differentiated stocks and externalities, so that the shadow value varies across grounds as well as over time. Another is to apply to the modern context that has implemented regulation; a harvest tax, tradable harvest rights, or restricted entry would require long-run dynamic evaluations.

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Navigating the Commons: Online Appendices

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September 6, 2026

Appendix A. Technical appendix

A.1. Details of dynamic choice probabilities and moment conditions

In the baseline analysis, I discretize the continuous capacity space into 41 grid levels, spaced at 200-ton increments, $\mathcal{K} = \{K^1 < K^2 < \cdots < K^J\}$ where $K^1 = 200$, $K^J = 8200$, and $J = 41$. I also discretize the productivity space into 7 grid levels, with transition probabilities computed directly from the data. The value functions are computed by backward induction. For years beyond $\overline{T}$, the terminal condition is the stationary-equilibrium value function described in Section 3.4; the recursion below then delivers V_t for every earlier year. Following Hotz and Miller (1993) and Gowrisankaran and Schmidt-Dengler (2025), I define the choice-specific value for selecting capacity K^j as

$$v_{it}^{j} = -C(K^j, K_{it}; \gamma) + \rho(1 - p_t)\, \mathbb{E}_t \left[V_{t+1}(s_{it+1}, \psi_{t+1}; \Theta) \,|\, s_{it}, \psi_t \right] \tag{A.1}$$

where j indexes the j-th grid of capacity space $\mathcal{K}$, and p_t is the wreck hazard from Section 3.3, because a staying firm reaches year $t+1$ only if it survives the end-of-year hazard.

From the incumbent's Bellman equation (12), the firm chooses K^j if and only if $v_{it}^j - I_{it}^j \epsilon_{it} \geq v_{it}^k - I_{it}^k \epsilon_{it}$, $\forall k \neq j$, where $I_{it}^j = K^j - (1-\delta)K_{it}$. Monotonicity of $I(\cdot, K_{it})$ implies that for $1 \leq j < k \leq J$, the firm prefers K^j to K^k if and only if

$$\epsilon_{it} \geq \frac{v_{it}^k - v_{it}^j}{I_{it}^k - I_{it}^j} = \epsilon_{it}(j,k)$$

where the last equality defines the "ϵ-cutoff" following Gowrisankaran and Schmidt-Dengler (2025). Intuitively, if the firm draws a higher (less favorable) cost shock than the cutoff, it chooses a smaller capacity. The firm chooses K^j if and only if $\underline{\epsilon}_{it}^j \leq \epsilon_{it} < \overline{\epsilon}_{it}^j$, where

$$\underline{\epsilon}_{it}^j = \begin{cases} \epsilon_{it}\big(j,\, j+1\big), & j < J, \\ -\infty, & j = J, \end{cases} \qquad \overline{\epsilon}_{it}^j = \begin{cases} +\infty, & j = 1, \\ \epsilon_{it}\big(j-1,\, j\big), & j > 1. \end{cases}$$

Then the probability of choosing capacity K^j, conditional on survival ($a_{it} > 0$), is given by

$$\Pr_t \left(a_{it} = K^j \,\middle|\, a_{it} > 0, s_{it}, \psi_t \right) = \Phi\left(\frac{\overline{\epsilon}_{it}^j}{\sigma} \right) - \Phi\left(\frac{\underline{\epsilon}_{it}^j}{\sigma} \right), \tag{A.2}$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

Another contribution of Gowrisankaran and Schmidt-Dengler (2025) is the computation of the conditional mean of the ϵ-shock without requiring numerical integration. By the properties of the truncated normal distribution, the conditional mean of the ϵ-shock is, $\forall j = 1, \ldots, J$,

$$\mathbb{E}\left[\epsilon_{it} \mid a_{it} = K^j, s_{it}, \psi_t\right] = \sigma \cdot \frac{\phi\left(\underline{\epsilon}_{it}^j/\sigma\right) - \phi\left(\overline{\epsilon}_{it}^j/\sigma\right)}{\Phi\left(\overline{\epsilon}_{it}^j/\sigma\right) - \Phi\left(\underline{\epsilon}_{it}^j/\sigma\right)}, \tag{A.3}$$

where $\phi(\cdot)$ is the standard normal probability density function. By using (A.1), (A.2), and (A.3), I define the expected continuation value as

$$V_{it}^{\text{cont}} = \sum_{j>0} \Pr_t\left(a_{it} = K^j \,\middle|\, a_{it} > 0, s_{it}, \psi_t\right) \left\{v_{it}^j - I_{it}^j\, \mathbb{E}\left[\epsilon_{it} \mid a_{it} = K^j, s_{it}, \psi_t\right]\right\} \tag{A.4}$$

From the expected continuation value and the logistic exit value draw $\zeta_{it} \sim \text{Logistic}(\bar{\zeta}, \sigma_L)$, the probability of exit is

$$\Pr_t\left(a_{it} = 0 \,\middle|\, s_{it}, \psi_t\right) = \Pr\left(\zeta_{it} \geq V_{it}^{\text{cont}}\right) = \frac{1}{1 + \exp\left[\left(V_{it}^{\text{cont}} - \bar{\zeta}\right)/\sigma_L\right]}. \tag{A.5}$$

Combining the period payoff (6), the exit probability (A.5), and the expected continuation value (A.4) yields a closed-form solution to the incumbent's Bellman equation:

$$V_t(s_{it}, \psi_t) = \Pi_t(s_{it}) + \mathbb{E}_\zeta\left[\max\left\{\zeta_{it},\, V_{it}^{\text{cont}}\right\}\right] = \Pi_t(s_{it}) + \sigma_L \ln\left(e^{\bar{\zeta}/\sigma_L} + e^{V_{it}^{\text{cont}}/\sigma_L}\right).$$

The log-sum-exp term is the expected value of the better of exiting and staying under the logistic draw; it exceeds $\max\{\bar{\zeta}, V_{it}^{\text{cont}}\}$ by the option value of the draw. The joint probability of survival and choosing capacity level K^j is given by

$$\Pr_t\left(a_{it} = K^j \,\middle|\, s_{it}, \psi_t\right) = \left(1 - \Pr_t\left\{a_{it} = 0 \,\middle|\, s_{it}, \psi_t\right\}\right) \Pr_t\left(a_{it} = K^j \,\middle|\, a_{it} > 0, s_{it}, \psi_t\right). \tag{A.6}$$

Finally, given the logistic entry cost draw $\kappa_{it} \sim \text{Logistic}(\bar{\kappa}, \sigma_L)$, the probability of entry is

$$\begin{aligned} \Pr_t\left(a_{it}^{\text{pe}} = 1 \,\middle|\, s_{it}, \psi_t\right) &= \Pr\left(\kappa_{it} \leq \rho\, \mathbb{E}_t[V_{t+1}(s_{it+1}, \psi_{t+1}) | s_{it}, \psi_t]\right) \\ &= \frac{1}{1 + \exp\left[-\left(\rho\, \mathbb{E}_t[V_{t+1}(s_{it+1}, \psi_{t+1}) | s_{it}, \psi_t] - \bar{\kappa}\right)/\sigma_L\right]}, \end{aligned} \tag{A.7}$$

where a potential entrant's payoff-relevant state is $s_{it} = (x^{\text{e}}, W_t, K_t, Q_t)$, with x^{e} the common entry state. A potential entrant is not yet at sea in year t, so its value is discounted at ρ alone and the wreck hazard first applies inside V_{t+1}.

Moment conditions. Equations (A.5), (A.2), and (A.7) give the three probabilities that the moment conditions compare with the observed decisions. I write them as

$$\chi_{it}(\Theta) = \Pr_t\left(a_{it} = 0 \,\middle|\, s_{it}, \psi_t\right), \qquad P^j_{it}(\Theta) = \Pr_t\left(a_{it} = K^j \,\middle|\, a_{it} > 0, s_{it}, \psi_t\right),$$
$$\chi^{\mathrm{e}}_t(\Theta) = \Pr_t\left(a^{\mathrm{pe}}_{it} = 1 \,\middle|\, s_{it}, \psi_t\right),$$

the probability of voluntary exit, the probability of choosing capacity K^j conditional on staying, and the probability of entry, which is common to all potential entrants in year t because they share the entry state x^{e}. Each probability depends on Θ through the equilibrium solve. The estimation sample of a phase consists of N firm-years of incumbents, the subset $\mathcal{S}$ of N_S firm-years that stayed and did not lose the entire fleet to a wreck, and the n_t entrants among the N^{pe}_t potential entrants of each year, with $N^{\mathrm{pe}} = \sum_t N^{\mathrm{pe}}_t$.[1]

A moment is the sample average of a generalized residual, a function of the observed action minus its conditional expectation under the model, multiplied by an instrument that depends only on the firm's state or the year. The exit residual is defined for every incumbent firm-year,

$$u^{\mathrm{x}}_{it}(\Theta) = \mathbb{1}\{a_{it} = 0\} - \chi_{it}(\Theta), \tag{A.8}$$

where the indicator equals one when the firm exits voluntarily. For a function f of the capacity chosen by a staying firm, the residual is

$$u^f_{it}(\Theta) = f(a_{it}) - \sum_{j=1}^{J} P^j_{it}(\Theta)\, f(K^j), \tag{A.9}$$

where a_{it} is the observed capacity choice and the sum is the model's expectation of f at the firm's state and year. Let k_{it} denote the grid index of current capacity, $K_{it} = K^{k_{it}}$, and $I^j_{it} = K^j - (1-\delta)K_{it}$ the net investment that reaches K^j, measured in thousand tons. The

[1] A firm-year in which a wreck destroys the entire fleet enters the exit moments as a stay and is excluded from the capacity moments, because the capacity the firm intended to operate is unobserved (Section 4.2). In the third phase, firm-years before 1856 are evaluated at the pre-petroleum solution and firm-years from 1856 on at the solution that starts at the news year.

baseline moment set uses six functions,

$$
\begin{aligned}
&f^{\text{inv}}(K^j) = \mathbb{1}\{j > k_{it}\}, && f^{\text{div}}(K^j) = \mathbb{1}\{j < k_{it}\}, \\
&f^{I+}(K^j) = I^j_{it}\ \mathbb{1}\{j > k_{it}\}, && f^{I-}(K^j) = I^j_{it}\ \mathbb{1}\{j < k_{it}\}, \\
&f^{\text{up}}(K^j) = \mathbb{1}\{j \geq k_{it} + 2\}, && f^{\text{dn}}(K^j) = \mathbb{1}\{j \leq k_{it} - 2\},
\end{aligned}
\tag{A.10}
$$

where the first pair records whether the firm invests or divests, the second pair how much, and the third pair whether it moves at least two grid steps (400 tons) in either direction. The entry residual is defined for each year of the phase,

$$
u^{\text{e}}_t(\Theta) = n_t - N^{\text{pe}}_t\, \chi^{\text{e}}_t(\Theta),
\tag{A.11}
$$

where n_t is the number of entrants observed in year t and N^{pe}_t is the pool of potential entrants from Section 3.1. The residual is the number of entrants minus the number the model expects. Under the model each residual has conditional mean zero given the state and the year at the true parameters, because the draws ζ_{it}, ϵ_{it}, and κ_{it} are independent of the state and the wreck is drawn after the decision. Any function of the state or the year is therefore a valid instrument. I use a constant, capacity $\tilde{K}_{it}$ and log productivity $\tilde{\omega}_{it}$, each standardized to mean zero and unit variance over the firm-years of the phase, and the year $\tilde{t}$, centered and scaled over the years of the phase.

The baseline moment set stacks 18 sample averages,

$$
\bar{g}(\Theta) = \begin{pmatrix}
\frac{1}{N} \sum_{(i,t)} u^{\text{x}}_{it}(\Theta) \left(1, \tilde{K}_{it}, \tilde{\omega}_{it}, \tilde{t}\right)' \\
\frac{1}{N_S} \sum_{(i,t)\in\mathcal{S}} u^{f}_{it}(\Theta) \left(1, \tilde{K}_{it}, \tilde{\omega}_{it}\right)', \quad f \in \{\text{inv}, \text{div}\} \\
\frac{1}{N_S} \sum_{(i,t)\in\mathcal{S}} u^{f}_{it}(\Theta) \left(1, \tilde{K}_{it}\right)', \quad f \in \{I+, I-\} \\
\frac{1}{N_S} \sum_{(i,t)\in\mathcal{S}} u^{f}_{it}(\Theta), \quad f \in \{\text{up}, \text{dn}\} \\
\frac{1}{N^{\text{pe}}} \sum_t u^{\text{e}}_t(\Theta) \left(1, \tilde{t}\right)'
\end{pmatrix},
\tag{A.12}
$$

four exit moments, twelve capacity moments, and two entry moments for the ten parameters $\bar{\kappa}$, $\bar{\zeta}$, σ_L, σ, and γ of a phase.[2]

[2] A larger set that adds the squared amounts invested and divested and the conditional variance of net investment, each with the constant as instrument, moves no estimate by more than [x] standard errors.

Estimator and weighting. The GMM estimator minimizes the weighted quadratic form in two steps,

$$\widehat{\Theta}_1 = \underset{\Theta}{\operatorname{argmin}}\ \bar{g}(\Theta)' W_1 \bar{g}(\Theta), \qquad \widehat{\Theta} = \underset{\Theta}{\operatorname{argmin}}\ \bar{g}(\Theta)' W_2 \bar{g}(\Theta), \tag{A.13}$$

where every evaluation of $\bar{g}(\Theta)$ re-solves the equilibrium of Section 3.4, $W_1 = \operatorname{diag}(1/v_1, \ldots, 1/v_{18})$ with v_k the sample variance of the data-side term of moment k (the observed function times its instrument) divided by the number of observations in its block, and $W_2 = \Omega_M(\widehat{\Theta}_1)^{-1}$ is the inverse of the model-implied covariance of $\bar{g}$ at the first-step estimate. The second step weights the moments by the inverse of their covariance when the actions are drawn from the model's own probabilities. That covariance is block diagonal across the three margins, because the exit, adjustment, and entry draws are independent,

$$\begin{aligned}
\Omega_M^{\mathrm{x}} &= \frac{1}{N^2} \sum_{(i,t)} \chi_{it}(1-\chi_{it})\, z_{it}^{\mathrm{x}} z_{it}^{\mathrm{x}\prime}, \\
\Omega_M^{\mathrm{K}} &= \frac{1}{N_S^2} \sum_{(i,t)\in\mathcal{S}} \Big[\sum_{j=1}^{J} P_{it}^{j}\, h_{it}(K^j)\, h_{it}(K^j)' - \bar{h}_{it}\bar{h}_{it}' \Big], \\
\Omega_M^{\mathrm{e}} &= \frac{\varphi}{(N^{\mathrm{pe}})^2} \sum_t N_t^{\mathrm{pe}}\, \chi_t^{\mathrm{e}}(1-\chi_t^{\mathrm{e}})\, z_t^{\mathrm{e}} z_t^{\mathrm{e}\prime},
\end{aligned} \tag{A.14}$$

where $z_{it}^{\mathrm{x}} = (1, \tilde{K}_{it}, \tilde{\omega}_{it}, \tilde{t})'$ and $z_t^{\mathrm{e}} = (1, \tilde{t})'$ are the exit and entry instruments, $h_{it}(K^j)$ stacks the twelve products of a capacity function in equation (A.10) with its instrument, $\bar{h}_{it} = \sum_j P_{it}^{j} h_{it}(K^j)$ is its conditional mean, and $\varphi \geq 1$ is the Pearson dispersion statistic of the yearly entry counts against the binomial, divided by its degrees of freedom and no less than one. The exit block is the binomial variance of the exit indicator, the capacity block is the variance of the instrumented functions over each staying firm's capacity distribution, and the entry block allows the yearly entry counts to be more dispersed than a binomial.[3]

Standard errors and overidentification. Let $D = \partial \bar{g}(\widehat{\Theta})/\partial \Theta'$ be the Jacobian of the moment vector with respect to the remaining parameters, computed by automatic differ-

[3] Before inversion each diagonal entry of Ω_M is bounded below at one thousandth of the corresponding data variance v_k, and the correlation matrix R of Ω_M is replaced by $(1-\alpha)R + \alpha I$ with $\alpha = [\alpha]$, so that the weighting matrix is well conditioned in the phases with few firms. The same inverse enters the J statistic below. Both steps minimize over the same bounded parameter space with a box-constrained quasi-Newton (L-BFGS) method, with gradients computed by forward-mode automatic differentiation through the equilibrium solve and the same convergence tolerances. The first step starts from the same parameter vector in every phase, $\bar{\kappa} = \bar{\zeta} = 100{,}000$, $\sigma = 100$, $\gamma_0^{\pm} = 10{,}000$, $\gamma_1^{\pm} = 100$, $\gamma_2^{\pm} = 0.5$, and $\sigma_L = 25{,}000$ in 1880 dollars, and the second step starts from $\widehat{\Theta}_1$.

entiation. The standard errors are the square roots of the diagonal of

$$\widehat{V} = \left(D'W_2D\right)^{-1} D'W_2\,\Omega_C\,W_2D\left(D'W_2D\right)^{-1}, \tag{A.15}$$

where Ω_C is the covariance of $\bar{g}$ with the exit and capacity contributions clustered jointly by firm and the entry block taken from equation (A.14). Clustering by firm allows a firm's exit and capacity residuals to be correlated across years and with each other.[4] The test of overidentifying restrictions is Hansen's J statistic,

$$J = \bar{g}(\widehat{\Theta})'\,\Omega_M(\widehat{\Theta})^{-1}\,\bar{g}(\widehat{\Theta}), \tag{A.16}$$

which is compared with the χ^2 distribution with degrees of freedom equal to 18 minus the number of free parameters. Table F3 reports each moment's t-ratio at $\widehat{\Theta}$. The standard deviation of moment k at $\widehat{\Theta}$ is the square root of the k-th diagonal element of

$$(I - \mathcal{P})\,\Omega_C\,(I - \mathcal{P})',$$

where $\mathcal{P} = D(D'W_2D)^{-1}D'W_2$ projects on the directions the estimated parameters fit. At a fixed parameter vector, such as the maximum likelihood estimates in the upper panel of that table, the standard deviation is the square root of the diagonal of Ω_C itself. Table F2 reports the GMM estimates beside the maximum likelihood estimates.

The moments condition on each firm's observed state path. Productivity is recovered for every firm-year in the first stage and discretized onto the grid, and entrants begin at the common state x^{e}, so no unobserved initial condition for the productivity process needs to be integrated out.

A.2. The planner's problem and whale shadow price

This section derives the shadow price of the whale stock from the planner's problem of Section 5.1, and then describes the first best computation.

The whale shadow price. The shadow price falls out of the first-order condition of problem (18) in the year-$(t+1)$ stock. Write $G_t := W_t + rW_t\left[1 - (W_t/W_1)^z\right] - Q_t$ for the stock that survives year t, so the constraint in (18) reads $G_t\phi^W_{t+1} - W_{t+1}$. The stock W_{t+1} enters

[4] The clustered blocks sum over firms the outer product of each firm's summed and centered contributions to $\bar{g}$, with the small-sample factor $n_f/(n_f - 1)$, where n_f is the number of firms in the phase. The entry block is taken from Ω_M because a phase has too few years for clustering by year.

the objective in exactly three places,

$$\rho^t \Lambda^W_{t+1}\big(-W_{t+1}\big), \qquad \rho^t\, \mathbb{W}_{t+1}, \qquad \rho^{t+1} \Lambda^W_{t+2}\big(G_{t+1}\phi^W_{t+2} - W_{t+2}\big),$$

the first because the constraint prices the stock the planner hands to year $t+1$, the second because that stock determines year-$(t+1)$ harvest and hence welfare, and the third because it also determines what survives into year $t+2$. Differentiating each and collecting,

$$-\rho^t\Lambda^W_{t+1} + \rho^t\frac{\partial \mathbb{W}_{t+1}}{\partial W_{t+1}} + \rho^{t+1}\Lambda^W_{t+2}\,\phi^W_{t+2}\,\frac{\partial G_{t+1}}{\partial W_{t+1}} = 0,$$

and dividing by ρ^t gives

$$\text{(A.17)} \qquad \Lambda^W_{t+1} = \frac{\partial \mathbb{W}_{t+1}}{\partial W_{t+1}} + \rho\,\Lambda^W_{t+2}\,\frac{\partial W_{t+2}}{\partial W_{t+1}}, \qquad \frac{\partial W_{t+2}}{\partial W_{t+1}} = \phi^W_{t+2}\,\frac{\partial G_{t+1}}{\partial W_{t+1}}, \qquad \Lambda^W_{T+1} = 0.$$

The multiplier is therefore measured in the dollars of the year whose stock it prices. Harvest in year t removes whales from the year-$(t+1)$ stock, so the marginal damage of harvesting in year t, in year-t dollars, is $\rho\,\Lambda^W_{t+1}$, the social cost of a landed whale that Section 5.1 reports.

The remaining derivative is the marginal welfare effect of an extra whale in year $t+1$. The stock enters $\mathbb{W}_{t+1}$ only through industry harvest, so

$$\frac{\partial \mathbb{W}_{t+1}}{\partial W_{t+1}} = \mathcal{P}_{t+1}(Q_{t+1})\,\Upsilon^W_{t+1}(f_{t+1}, W_{t+1}),$$

where $\Upsilon^W_{t+1}(f_{t+1}, W_{t+1}) := \sum_z f_{t+1}(z)\,\partial\mathcal{H}_{t+1}(z, W_{t+1}, K_{t+1})/\partial W_{t+1}$ is the marginal effect of the whale stock on industry harvest, that is, the stock externality. Substituting it into (A.17) gives

$$\Lambda^W_{t+1} = \mathcal{P}_{t+1}(Q_{t+1})\,\Upsilon^W_{t+1}(f_{t+1}, W_{t+1}) + \rho\,\Lambda^W_{t+2}\,\frac{\partial W_{t+2}}{\partial W_{t+1}}.$$

Iterating forward gives the whale shadow price,

$$\text{(A.18)} \qquad \Lambda^W_{t+1} = \sum_{\ell=t+1}^{T} \rho^{\ell-t-1}\left(\prod_{m=t+1}^{\ell-1}\frac{\partial W_{m+1}}{\partial W_m}\right)\mathcal{P}_\ell(Q_\ell)\Upsilon^W_\ell(f_\ell, W_\ell),$$

the discounted value of every future harvest $\mathcal{P}_\ell\Upsilon^W_\ell$ that one more whale would have yielded, propagated forward through the stock dynamics $\partial W_{m+1}/\partial W_m$.

Computing the first best. [IT'S AWKWARD TO INTRODUCE w "node" INDEXES SUDDENLY. MAYBE NEED TO DROP THAT. ALSO, IS THE NOTATION $\mathcal{V}_t(f_t, W)$ NECESSARY?] The planner chooses rules that are functions of the year and the firm's own state x. Firm states move through capacity choice, the productivity process, and the wreck hazard. Write $\pi_t(w)$ for the probability that the stock is at grid point W^w in year t, and $\Pr_t(j \mid w)$ for the probability of moving from W^w to W^j. Then π_1 is a point mass at the observed stock in 1815 and

$$\pi_{t+1}(j) = \sum_w \Pr_t(j \mid w)\, \pi_t(w). \tag{A.19}$$

Harvest, price, and consumer surplus are exact functions of the node, with $Q_t(w) = \sum_x f_t(x)\mathcal{H}_t(x, W^w, K_t)$, so expected welfare is the π_t-average of the market flow plus the adjustment, entry, and exit terms, which do not depend on the stock.

Write $\mathcal{V}_t(f_t, W)$ for the planner's continuation value from year t onward when the stock is W, evaluated under the candidate rules. On the chain the shadow price is then a difference across nodes rather than a derivative in a scalar. Landing one more whale at node w shifts that node's transition row toward lower successor stocks, and the shadow price is what the shift is worth at those continuation values,

$$\Lambda^W_{t+1}(w) = -\sum_j \frac{\partial \Pr_t(j \mid w)}{\partial Q_t(w)}\, \mathcal{V}_{t+1}(f_{t+1}, W^j). \tag{A.20}$$

Because the transition probabilities sum to one at every node, the terms of $\partial \Pr_t(j \mid w)/\partial Q_t(w)$ sum to zero, so (A.20) prices only the movement of probability mass between continuation values and is unchanged by anything added to $\mathcal{V}_{t+1}$ at every node. The stock-independent adjustment, entry, and exit terms therefore drop out of the shadow price, exactly as they drop out of (A.17).

The planner's flow payoff for a firm at state x is the π_t-average of

$$\big[\mathcal{P}_t(Q_t(w)) - \rho\, \Lambda^W_{t+1}(w)\big]\big(\mathcal{H}_t(x, W^w, K_t) + K(x)\, \Upsilon^K_t(w)\big) - w^L_t L(K(x)),$$

where $\Upsilon^K_t(w) := \beta^K Q_t(w)/K_t$ is the marginal effect of industry capacity on harvest. The firm's own harvest and the harvest its capacity crowds out from the rest of the fleet are both valued at the social price $\mathcal{P}_t - \rho\,\Lambda^W_{t+1}$, the market price net of the stock damage. Exit, capacity adjustment, and entry then follow the same cost and value draws $(\zeta_{it}, \epsilon_{it}, \kappa_{it})$ that firms face.

I solve for a fixed point in the rules. Given candidate rules, the firm distribution and aggregate capacity are propagated forward, the chain (A.19) and the node objects are built from them, the value function and (A.20) run backward, and the implied rules are mixed into the candidate. The iteration repeats until the rules stop moving. Because the fleet is deterministic under year-only rules, the converged profile maximizes expected welfare over the class, up to the discretization of the stock grid.

The planner faces the same whale-stock uncertainty as firms, at the same calibrated σ_W, and holds the same information about the petroleum discovery. I therefore solve it in two stages. The first stage optimizes over the whole horizon on the pre-petroleum demand curve, which is the future firms and planner alike believed in. The second stage freezes the first stage's rules for every year before the petroleum discovery and re-optimizes from the that year onward against the realized demand path.

Appendix B. Agents' supervision

The following examples are borrowed from Chapter 10 of Davis, Gallman, and Gleiter (2007) and Chapter 1 of Nicholas (2019).

B.1. Planning by agent

A formal statement of the main outlines of the plan was usually given to the captain, as seen in the following passage from a letter dated 1 November 1834 from *Agent* Charles W. Morgan to *Captain* Reuben Russell, 2d.

> The Bark being now ready for sea, as agent I have to advise you that she is bound on a whaling voyage to the Pacific Ocean-That she is fitted for thirty months-and that we wish you to cruise for sperm whales for 20 to 24 months and if not then full, fill up with whale Oil-we leave to your judgment the cruising ground on the Pacific though we would recommend the neighborhood of New Zealand, where both right & sperm whales are to be taken, and it would be well especially towards the end of the voyage to be where right whales could be taken. (Morgan Collection)

The following letter from agent Charles W. Morgan to a young captain, George H. Dexter, shows that agents also provided advice and encouragement.

> As you have now taken the responsible station of Master of a Ship and are a young man you will permit me to offer some advice. The greatest difficulty I have observed with young Masters, is either too great indulgence or too great severity towards their crew. Discipline must be effectual, be administered with a steady hand especially among Sailors, and there is no station which requires more guard over the temper, than that of a master of a Ship. And on your first voyage depends in a great measure your future success in life. Let me then beg of you to keep a strict watch over the moral conduct of your crew, never permit your authority to be abused or set at naught, but at the same time never to use undue severity yourself or permit it in your officers. I have a real confidence in you and I trust you will not feel hurt at my giving advice which I feel it my duty to offer you. (Morgan Collection)

B.2. Contact between agent and captain during the voyage

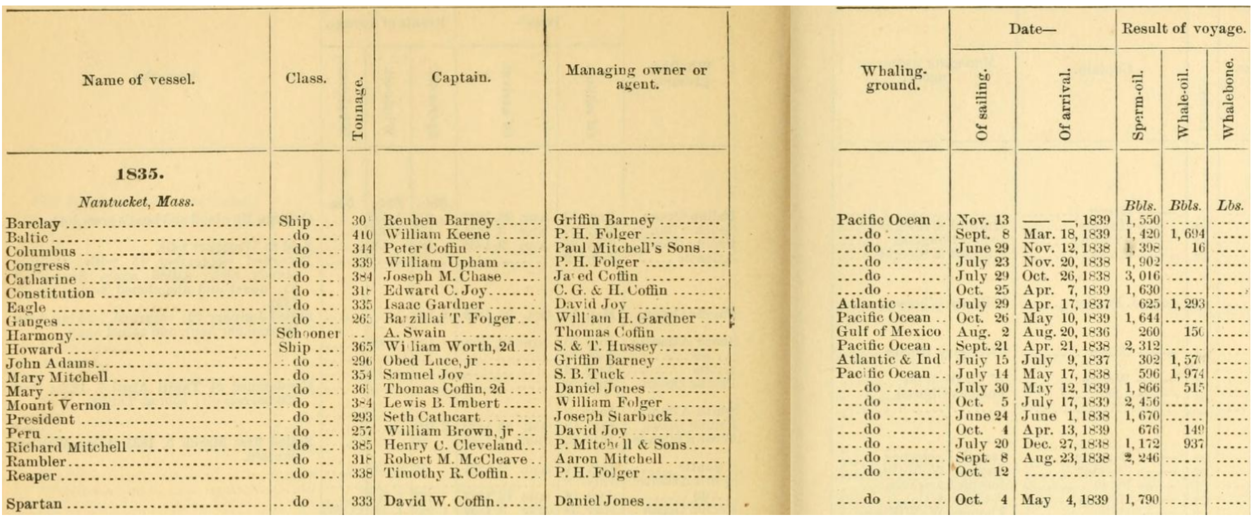
A glimpse of the exchanges between the captain and agent can be obtained from the letter books of whaling agents. For instance, in February 1858, Matthew Howland wrote to one of his captains, Philip Howland, advising him on his recent performance:

> 25 months out with 600 bbl sperm & 130 whale is rather low, but I am in hopes that you will come up now and be equal to any of them according to time out—I shall expect to hear of you into Talcuahana in March with from 800 to 1000 bbls of sperm oil on board. (Howland Collection)

Appendix C. Details of data

This section describes the data sources and modifications.

C.1. *Voyage Database*



Name of vessel.	Class.	Tonnage.	Captain.	Managing owner or agent.	Whaling-ground.	Date— Of sailing.	Date— Of arrival.	Result of voyage. Sperm-oil.	Result of voyage. Whale-oil.	Result of voyage. Whalebone.
1835.										
Nantucket, Mass.								*Bbls.*	*Bbls.*	*Lbs.*
Barclay	Ship	30	Reuben Barney	Griffin Barney	Pacific Ocean	Nov. 13	—— —, 1839	1,550	......	......
Baltic	do	410	William Keene	P. H. Folger	do	Sept. 8	Mar. 18, 1839	1,420	1,694	......
Columbus	do	314	Peter Coffin	Paul Mitchell's Sons	do	June 29	Nov. 12, 1838	1,398	16	......
Congress	do	339	William Upham	P. H. Folger	do	July 23	Nov. 20, 1838	1,902	......	......
Catharine	do	384	Joseph M. Chase	Jared Coffin	do	July 29	Oct. 26, 1838	3,016	......	......
Constitution	do	318	Edward C. Joy	C. G. & H. Coffin	do	Oct. 25	Apr. 7, 1839	1,630	......	......
Eagle	do	335	Isaac Gardner	David Joy	Atlantic	July 29	Apr. 17, 1837	625	1,293	......
Ganges	do	265	Barzillai T. Folger	William H. Gardner	Pacific Ocean	Oct. 26	May 10, 1839	1,644	......	......
Harmony	Schooner	..	A. Swain	Thomas Coffin	Gulf of Mexico	Aug. 2	Aug. 20, 1836	260	150	......
Howard	Ship	365	William Worth, 2d	S. & T. Hussey	Pacific Ocean	Sept. 21	Apr. 21, 1838	2,312	......	......
John Adams	do	296	Obed Luce, jr	Griffin Barney	Atlantic & Ind	July 15	July 9, 1837	302	1,570	......
Mary Mitchell	do	354	Samuel Joy	S. B. Tuck	Pacific Ocean	July 14	May 17, 1838	596	1,974	......
Mary	do	361	Thomas Coffin, 2d	Daniel Jones	do	July 30	May 12, 1839	1,866	515	......
Mount Vernon	do	384	Lewis B. Imbert	William Folger	do	Oct. 5	July 17, 1839	2,456	......	......
President	do	293	Seth Cathcart	Joseph Starbuck	do	June 24	June 1, 1838	1,670	......	......
Peru	do	257	William Brown, jr	David Joy	do	Oct. 4	Apr. 13, 1839	676	149	......
Richard Mitchell	do	385	Henry C. Cleveland	P. Mitchell & Sons	do	July 20	Dec. 27, 1838	1,172	937	......
Rambler	do	318	Robert M. McCleave	Aaron Mitchell	do	Sept. 8	Aug. 23, 1838	2,246	......	......
Reaper	do	338	Timothy R. Coffin	P. H. Folger	do	Oct. 12	....	......	......	......
Spartan	do	333	David W. Coffin	Daniel Jones	do	Oct. 4	May 4, 1839	1,790	......	......

FIGURE C1. Sample of Starbuck (1878)

Source: Starbuck, A. 1878. *History of the American whale fishery from its earliest inception to the year 1876*.

The American Whaling Voyage database includes information about all known US whaling voyages from the 1700s to the 1920s. Voyages have been defined based on customs house records, following Starbuck (1878), with each departure and subsequent return to the port of origin constituting a single voyage. Figure C1 provides an example page from Starbuck (1878). A basic suite of information is included for most voyages, along with additional information on the ship's capacity and rig, declared destination, and amount of whale products. The database is digitized and provided by Mystic Seaport Museum and New Bedford Whaling Museum (https://whalinghistory.org/).

C.2. Recovering missing agents: vessel continuity and *Ship Registers*

The *Voyage Database* lists no whaling agent for 5,182 of the 15,707 voyages (33 percent), and the gaps cluster before the 1830s. I recover these agents in two steps: vessel continuity and ship registers.

The first step exploits vessel continuity. A whaling agent is the firm that outfits and manages a voyage and sells its catch; the agent runs the vessel, so it typically changes when the vessel changes hands, not from one year to the next. When a voyage lists no agent

6 ABIGAIL, ship, of New Bedford. Registered (23) July 18, 1821 - permanent. Built at Amesbury in 1810. 309 75/95 tons; length 97 ft., breadth 27 ft., depth 13 ft. 6 in. Master: Dennis Covell, Owners: Benjamin Rodman, merchant, Andrew Robeson, David Coffin, Dennis Covell, Elisha Dunbar, New Bedford. Two decks, three masts, square stern, no galleries, a billethead. Previously registered at Newburyport Mar. 21, 1821.

7 Ship, of New Bedford. Re-registered (38) Nov. 17, 1823 - permanent. Master: Hezekiah B. Gardner. Owners: Benjamin Rodman, Andrew Robeson, David Coffin, Elisha Dunbar, merchants, New Bedford.

8 Ship, of New Bedford. Re-registered (47) Dec. 22, 1825 - permanent. Master: Stephen Potter. Owners: Benjamin Rodman, merchant, Charles W. Morgan, David Coffin, Elisha Dunbar, New Bedford.

9 Ship, of New Bedford. Re-registered (73) Nov. 19, 1831 - permanent. Master: Benjamin Clark. Owners: Benjamin Rodman, Charles W. Morgan, David Coffin, Elisha Dunbar, Benjamin Clark, New Bedford.

10 Ship, of New Bedford. Re-registered (43) June 13, 1835 - permanent. Owners: Charles W. Morgan, William R. Rodman, David Coffin, Benjamin Clark, Elisha Dunbar, New Bedford.

FIGURE C2. A sample of ship registers of New Bedford: Vessel *Abigail*

Source: Ship registers of New Bedford from HathiTrust (https://www.hathitrust.org/).

but the same vessel has a named agent on a voyage within two years, I assign the missing voyage that agent. This step recovers 1,149 voyages. Because it copies agents already in the database, it adds voyage-years to existing firms rather than creating new ones.[5]

The second step handles vessels with no named agent anywhere in the database. For these I read the port *Ship Registers*—chiefly New Bedford, together with Boston, Salem, Providence, Newport, Bristol–Warren, Plymouth, and Newburyport. I locate the registration in force at the departure year, confirm the vessel by its tonnage, and take the first-named owner as the agent, except where the register records ownership shares or a named managing firm, in which case I use the majority owner or that firm. This adds several hundred further voyages, some run by firms otherwise absent from the database.

Together these steps raise agent coverage from 10,525 voyages—about two-thirds of the total—to 12,612, or 80 percent of all U.S. voyages.

[5]Figure C2 provides a useful test example, using the vessel *Abigail* from the ship registers of New Bedford. The voyage database records the *Abigail*'s agent as Benjamin Rodman in 1825 and Charles W. Morgan from 1835 onward, but leaves the 1821 voyage blank. Continuity assigns the 1825 agent, Rodman, to the adjacent 1821 voyage. The registers corroborate the assignment—Rodman heads the owner list in both 1821 and 1825. To gauge it more systematically, I hold out the agent of each voyage that does record one and predict it from the same vessel: this rule recovers the true agent for 83 percent of the held-out voyages, and the correct surname for 88 percent.

C.3. Modifications

For the purpose of empirical analysis, the output quantities (in barrels of sperm oil and whale oil, and pounds of whalebone) are converted into the number of whales harvested. Following Scammon (1874), this paper assumes that an average sperm whale taken yielded 25 barrels of sperm oil and an average baleen whale taken yielded 60 barrels of whale oil. Historical accounts also document that around 10% of whales escaped but subsequently died from their wounds. Therefore I relate the barrels of oil observed in the data to the number of whales harvested in the following way:

$$Q_{it}^{SW} = \frac{Q_{it}^{\text{soil}}}{25} + 0.1 \times \frac{Q_{it}^{\text{soil}}}{25}$$
$$Q_{it}^{BWO} = \frac{Q_{it}^{\text{woil}}}{60} + 0.1 \times \frac{Q_{it}^{\text{woil}}}{60}$$

where Q_{it}^{SW} is the number of sperm whales harvested by firm i in year t, Q_{it}^{soil} is the barrels of sperm oil, Q_{it}^{BWO} is the number of baleen whales harvested to make whale oil, and Q_{it}^{woil} is the barrels of whale oil. From the logbook database,[6] I assume that an average baleen whale taken yielded 500 pounds of whalebone.

$$Q_{it}^{BWB} = \frac{Q_{it}^{\text{bone}}}{500} + 0.1 \times \frac{Q_{it}^{\text{bone}}}{500}$$

where Q_{it}^{BWB} is the number of baleen whales harvested to make whalebone and Q_{it}^{bone} is the pounds of whalebone. Then, the number of baleen whales harvested, Q_{it}^{BW}, is determined as follows:

$$Q_{it}^{BW} = \max\left\{Q_{it}^{BWO}, Q_{it}^{BWB}\right\}$$

Finally, the whaling output by firm i in year t is defined as the total number of whales harvested:

$$Q_{it} = Q_{it}^{SW} + Q_{it}^{BW}$$

[6] The American Offshore Whaling Log database comprises information from 1,381 logbooks documenting US whaling voyages spanning 1784 to 1920. This dataset was extracted from the original whaling logbooks during three distinct scientific research projects. The first initiative was led by Lt. Cmdr. Matthew Fontaine Maury in the 1850s, the second by Charles Haskins Townsend in the 1930s, and the third by a team from the Census of Marine Life project (CoML, www.coml.org), led by Tim Denis Smith from 2000 to 2010. The CoML team assembled the Maury and Townsend data from archival sources. The data file encompasses 466,134 records organized in a standardized format, including voyage ID, coordinates (latitude/longitude), date, whale encounter, species, harpooned, place, and so on.

Appendix D. Estimation strategy for a nonstationary global whaling market

I use a full-solution method instead of conditional choice probability approaches for two reasons.

The first reason relates to counterfactual analysis. This paper evaluates firms' behavioral responses to regulatory, environmental, and economic changes in exploiting shared resources. Such analysis requires counterfactual equilibrium decision rules obtained by re-solving the model under alternative conditions. For example, imposing a tax on whaling output raises costs and alters firms' equilibrium behavior. To assess these changes, the model must be solved again. A full-solution approach ensures consistency between estimation and counterfactual exercises, enabling direct and reliable comparisons of model outcomes.

The second reason relates to estimation. The global whaling industry presents significant empirical challenges due to the nonstationary evolution of economic fundamentals.[7] Figures F1, 1, and 2 illustrate the dynamics of American whaling, including demand shifts, whale stock transition, and technological advancements. Since this paper examines whaling firms' entry, exit, and investment decisions in such a nonstationary environment, the firms' policy functions must adapt over time. In essence, time becomes an additional state variable, making policy functions explicitly time-dependent, as detailed in Section 3.3.

The cross-sectional and time-series characteristics of the data reduce the effective sample size for every year, complicating the application of two-step conditional choice probability methods (e.g., Aguirregabiria and Mira 2007; Bajari, Benkard, and Levin 2007; Pakes, Ostrovsky, and Berry 2007; Pesendorfer and Schmidt-Dengler 2008). First-step nonparametric estimates of conditional choice probabilities become imprecise when only a few or no choices are observed in certain states. The model has to fill in these blanks, and hence this paper takes a full-solution approach instead, as in Benkard (2004), Goettler and Gordon (2011), Igami (2017), and Elliott (2024).

[7]Nonstationarity is particularly important in resource industries. For example, the American whaling industry experienced rapid growth and decline over a century due to changes in whale stocks, alternative resource availability, and technological advancements. Similar trends can be observed in modern energy sectors, such as cost reductions and increasing demand for solar panels, electric vehicles, and other renewables.

Appendix E. Fit of the estimated model

I evaluate model fit by comparing observed behavior in the data with predictions from the estimated model. Figure 3 compares the data (solid line) with model-implied paths (dashed line) for total whale harvest (Panel A) and total whaling-vessel capacity (Panel B). The model paths are averages across 200 simulations under the estimated parameters, with shaded bands indicating 95 percent confidence intervals. The figure shows that the model reproduces the industry's rise and fall over nine decades, including the expansion through the mid-1840s and the subsequent long decline.

The model does, however, lag the industry's collapse in the late 1850s and early 1860s. Harvests in the data fall by half between 1853 and 1861; the model takes until 1864 to shed the same share, so the simulated path runs above the data from the mid-1850s through 1864. Two forces concentrated in this window lie outside the model. First, catch per ton fell sharply in the late 1850s as the most productive hunting grounds were depleted and voyages lengthened. But the production function ties harvests to the aggregate capacity, whale stock and a smooth trend (technology growth), so it spreads this localized depletion over the following decade. Second, the estimated demand curve does not capture that camphene and coal-derived kerosene that were already displacing whale-based illuminants before 1859. Both channels hold simulated returns, and with them the simulated total capacity and harvest, higher than the data on the eve of the Civil War.

Appendix F. Additional tables and figures

TABLE F1. Demand-side statistics for the American whaling industry

Variable	Unit	Mean	SD	Min	Max
Whales demanded	Number	4206.40	3912.06	23.21	12249.07
Whale price	1880 dollars per whale	796.32	322.34	298.09	1984.68
US real GDP per capita	1996 dollars	2434.03	1103.68	1249.00	5357.00
Petroleum price	1880 dollars	0.05	0.06	0.01	0.41
Year		1856.50	30.74	1804.00	1909.00

Notes: The dataset comprises 106 observations. One exception is petroleum price data, which dates back to 1859, the year of petroleum discovery.

TABLE F2. Dynamic parameters by maximum likelihood and by GMM

	Unit	1815-29	1830-44	1845-59	1860-74	1875-89	1890-1905
Panel A. Maximum likelihood estimates							
Mean of entry costs: $\bar{\kappa}$	\$000	315.94	390.23	874.44	1078.08	1168.61	1038.46
		(38.53)	(22.54)	(60.33)	(235.86)	(137.21)	(213.58)
Mean of exit scrap values: $\bar{\zeta}$	\$000	188.42	267.45	671.82	868.76	964.79	715.55
		(37.59)	(25.22)	(51.39)	(204.92)	(129.08)	(151.63)
Scale parameter: σ_L	\$000	73.53	64.27	89.81	82.06	116.26	143.91
		(17.99)	(9.54)	(10.00)	(14.33)	(30.61)	(52.24)
Investment fixed: γ_0^+	\$000	5.92	8.84	11.04	5.89	0.00	23.91
		(3.47)	(1.58)	(1.77)	(3.45)	(9.28)	(20.23)
Investment linear: γ_1^+	\$/ton	343.37	284.07	308.50	461.89	858.99	1000.00
		(32.90)	(17.67)	(16.13)	(92.55)	(148.53)	(204.78)
Investment convex: γ_2^+	\$/ton^2	0.33	0.25	0.29	0.33	0.65	1.00
		(0.06)	(0.03)	(0.03)	(0.11)	(0.16)	(0.26)
Divestment fixed: γ_0^-	\$000	13.28	22.80	20.79	28.65	68.61	68.91
		(5.93)	(3.41)	(2.61)	(10.32)	(19.10)	(23.53)
Divestment linear: γ_1^-	\$/ton	0.00	0.00	65.92	121.18	0.00	0.00
		(63.02)	(27.45)	(20.49)	(41.92)	(123.67)	(171.45)
Divestment convex: γ_2^-	\$/ton^2	0.33	0.31	0.26	0.40	1.10	0.64
		(0.09)	(0.04)	(0.03)	(0.14)	(0.28)	(0.19)
SD of marginal adj. costs: σ	\$/ton	224.94	221.38	206.06	299.18	736.78	749.21
		(43.21)	(25.53)	(22.28)	(103.17)	(160.52)	(157.67)
Log likelihood		-1440.0	-4313.5	-4482.4	-2240.0	-1071.3	-464.3
Number of observations		1097	3266	3511	1690	751	371
Panel B. GMM estimates, primary moment set, generic cold start (21 moments)							
Mean of entry costs: $\bar{\kappa}$	\$000				1206.63		
					(127.58)		
Mean of exit scrap values: $\bar{\zeta}$	\$000				1027.66		
					(118.24)		
Scale parameter: σ_L	\$000				70.43		
					(10.34)		
Investment fixed: γ_0^+	\$000				0.44		
					(2.73)		
Investment linear: γ_1^+	\$/ton				382.73		
					(56.34)		
Investment convex: γ_2^+	\$/ton^2				0.20		
					(0.05)		
Divestment fixed: γ_0^-	\$000				30.60		
					(6.55)		
Divestment linear: γ_1^-	\$/ton				182.49		
					(25.49)		
Divestment convex: γ_2^-	\$/ton^2				0.37		
					(0.08)		
SD of marginal adj. costs: σ	\$/ton				199.57		
					(39.44)		
Hansen J [d.f.]					545.1 [11]		
J p-value					0.000		
Log likelihood at GMM estimates					-2271.1		
Panel C. GMM estimates, primary moment set, MLE campaign start point (21 moments)							
Mean of entry costs: $\bar{\kappa}$	\$000						1127.54

TABLE F3. Fit of the baseline moments

[Table to be generated by the `estimation_table_gmm` section.]

Note: This table reports the t-ratio of each of the 18 sample moments in equation (A.12), the sample moment divided by its standard deviation, one column per phase. The upper panel evaluates the moments at the maximum likelihood estimates of Table F2, holding the parameters fixed, with the standard deviation from the covariance clustered by firm. The lower panel evaluates them at the GMM estimates of Table 5, with the standard deviation adjusted for the part of each moment that the estimated parameters absorb (Online Appendix A.1). $\tilde{K}$, $\tilde{\omega}$, and $\tilde{t}$ denote the standardized capacity, log productivity, and year instruments, I^+ and I^- the amounts invested and divested in thousand tons, and a step is one 200-ton grid level. The J statistic in Table F2 is the joint test. [The largest t-ratio at the GMM estimates is [xx], on the [moment] moment in [phase].]

TABLE F4. Dynamic cost estimates under alternative stock-belief uncertainty σ_W

	Unit	0.10	0.20	0.30
Dynamic cost estimates by σ_W (phase: 1845–59)				
Mean of entry costs: $\bar{\kappa}$	\$000	830.93	885.30	922.50
		(88.98)	(93.19)	(95.83)
Mean of exit scrap values: $\bar{\zeta}$	\$000	158.37	168.02	174.67
		(13.72)	(14.20)	(14.46)
SD of marginal adj. costs: σ	\$/ton	212.46	215.08	217.79
		(16.90)	(17.75)	(18.34)
Investment fixed: γ_0^+	\$000	11.72	11.89	12.07
		(1.67)	(1.71)	(1.74)
Investment linear: γ_1^+	\$/ton	314.32	318.41	321.33
		(16.09)	(16.31)	(16.52)
Investment convex: γ_2^+	\$/ton^2	0.32	0.32	0.33
		(0.03)	(0.03)	(0.03)
Divestment fixed: γ_0^-	\$000	19.77	20.07	20.36
		(2.15)	(2.22)	(2.28)
Divestment linear: γ_1^-	\$/ton^2	82.57	83.48	83.41
		(17.46)	(18.07)	(18.51)
Divestment convex: γ_2^-	\$/ton	0.27	0.27	0.28
		(0.03)	(0.03)	(0.03)
Log likelihood		4500.4	4504.2	4506.6

Note: σ_W is the standard deviation, in log points, of firms' beliefs about next year's whale stock; the baseline sets $\sigma_W = 0.10$, interpreted as belief dispersion rather than realized variation in whale abundance (Section 4.2). Each column re-estimates the dynamic cost parameters holding σ_W fixed at the stated value, with all other primitives at their baseline values. Estimates are for the 1845–59 phase; standard errors are in parentheses. Costs rise with σ_W because the harvesting technology is convex in the whale stock ($\beta^W > 1$), so wider belief dispersion raises expected costs through Jensen's inequality. Baseline estimates are in Table 5.

TABLE F5. Sensitivity to the post-1859 demand elasticity

	Post-1859 price elasticity α_p		
	−2.0	−4.46 (estimated)	−5.0
Laissez-faire ($M)	–	46.37	–
First best ($M)	–	60.65	–
Loss ($M)	–	14.28	–
Share of first best	–	24%	–
Peak capacity vs. first best	–	3.6×	–

Note: The price elasticity α_p is estimated on the pre-petroleum sample and imposed on the post-1859 period, where the instruments are weak. This table re-estimates the post-1859 demand shifters and re-solves the laissez-faire and first-best allocations at $\alpha_p = -2.0$ and -5.0, holding all other primitives at their baseline values. Welfare is in millions of 1880 dollars, discounted to 1815.

TABLE F6. Sensitivity to the production-function capacity elasticity β_k

		Baseline	$\hat{\beta}_k$ + 1SE	$\hat{\beta}_k$ + 2SE
Panel A. Production function				
Own capacity β_k		1.049	1.056	1.063
Productivity persistence ρ		0.159	0.424	0.555
Congestion β_K		-0.067	-0.064	-0.062
Whale stock β_w		5.136	5.294	5.436
Time trend β_t		0.0133	0.0137	0.0140
Panel B. Dynamic cost estimates (phase: 1845–59)				
Mean of entry costs: $\bar{\kappa}$	\$000	885.30	881.42	886.30
		(93.19)	(93.48)	(94.26)
Mean of exit scrap values: $\bar{\zeta}$	\$000	168.02	171.22	171.81
		(14.20)	(14.45)	(14.55)
SD of marginal adj. costs: σ	\$/ton	215.08	224.03	231.28
		(17.75)	(18.51)	(19.30)
Investment fixed: γ_0^+	\$000	11.89	12.40	12.79
		(1.71)	(1.78)	(1.84)
Investment linear: γ_1^+	\$/ton	318.41	326.83	330.91
		(16.31)	(16.91)	(17.45)
Investment convex: γ_2^+	\$/ton^2	0.32	0.34	0.35
		(0.03)	(0.03)	(0.03)
Divestment fixed: γ_0^-	\$000	20.07	20.88	21.57
		(2.22)	(2.32)	(2.41)
Divestment linear: γ_1^-	\$/ton^2	83.48	82.24	78.23
		(18.07)	(18.84)	(19.50)
Divestment convex: γ_2^-	\$/ton	0.27	0.28	0.29
		(0.03)	(0.03)	(0.03)
Log likelihood		4504.2	4505.4	4508.2

Note: The capacity elasticity β_k governs how sharply the productivity shock selects which firms exit. I raise $\hat{\beta}_k$ by one and two first-stage standard errors, re-estimate the production function (Panel A) and the dynamic cost parameters (Panel B), and hold all other primitives at their baseline. Panel B reports the 1845–59 phase; standard errors are in parentheses. The dynamic cost estimates are broadly stable across the perturbations. Baseline estimates are in Table 5.

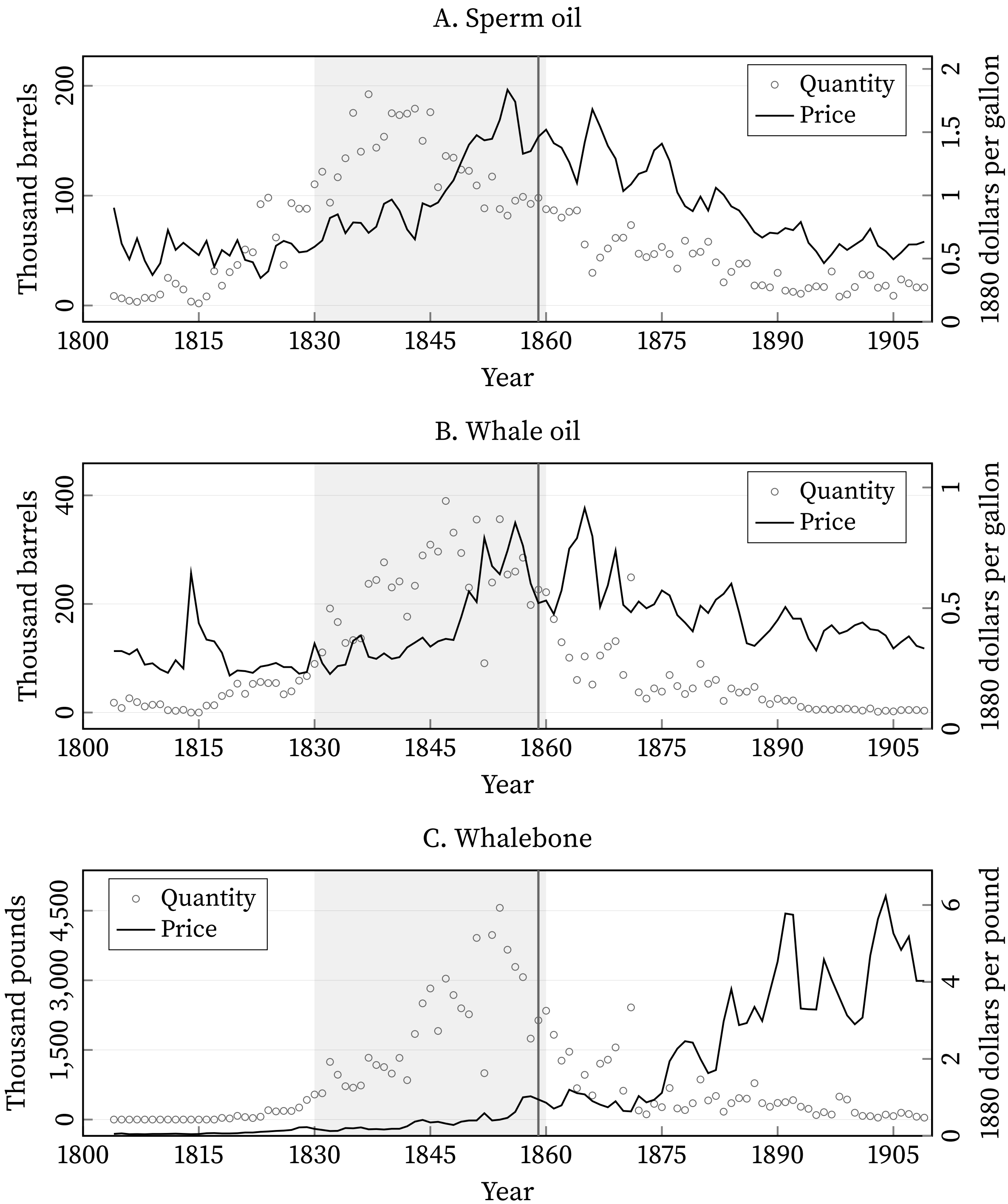


FIGURE F1. Quantities and prices for whaling products

Note: The term “Golden Age” refers to the period when the American whaling industry was at its peak, from the 1830s to the 1850s. The year of petroleum discovery in 1859 is indicated by the vertical red line.

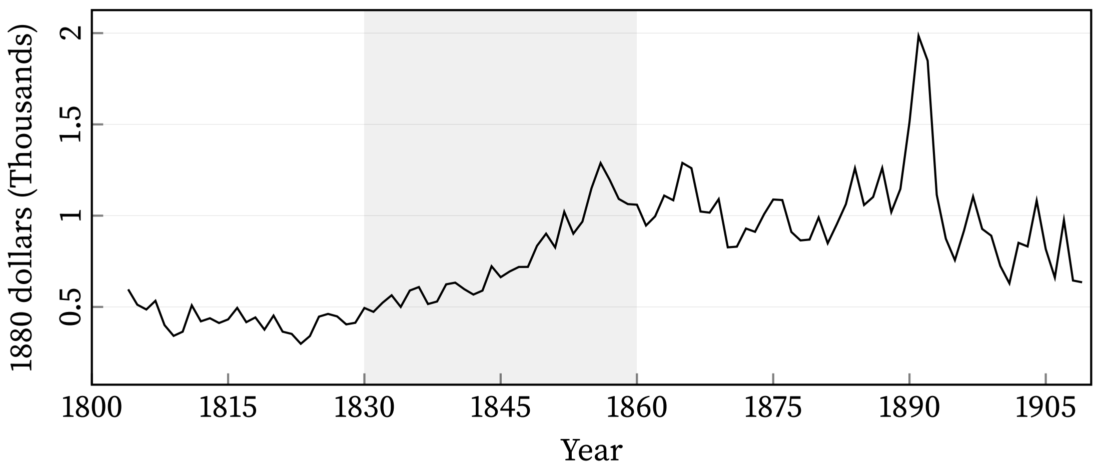


FIGURE F2. Whale price

Note: The prices of whaling products come from Tower (1907) and Davis, Gallman, and Gleiter (2007). I calculate the price *per whale* as total output value divided by total whales harvested each year. The output value equals sperm oil quantity × price + whale oil quantity × price + whalebone quantity × price. I compute total whales harvested by summing catches across all voyages.

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